

如何控制、校正超導 量子位元？

王培儒

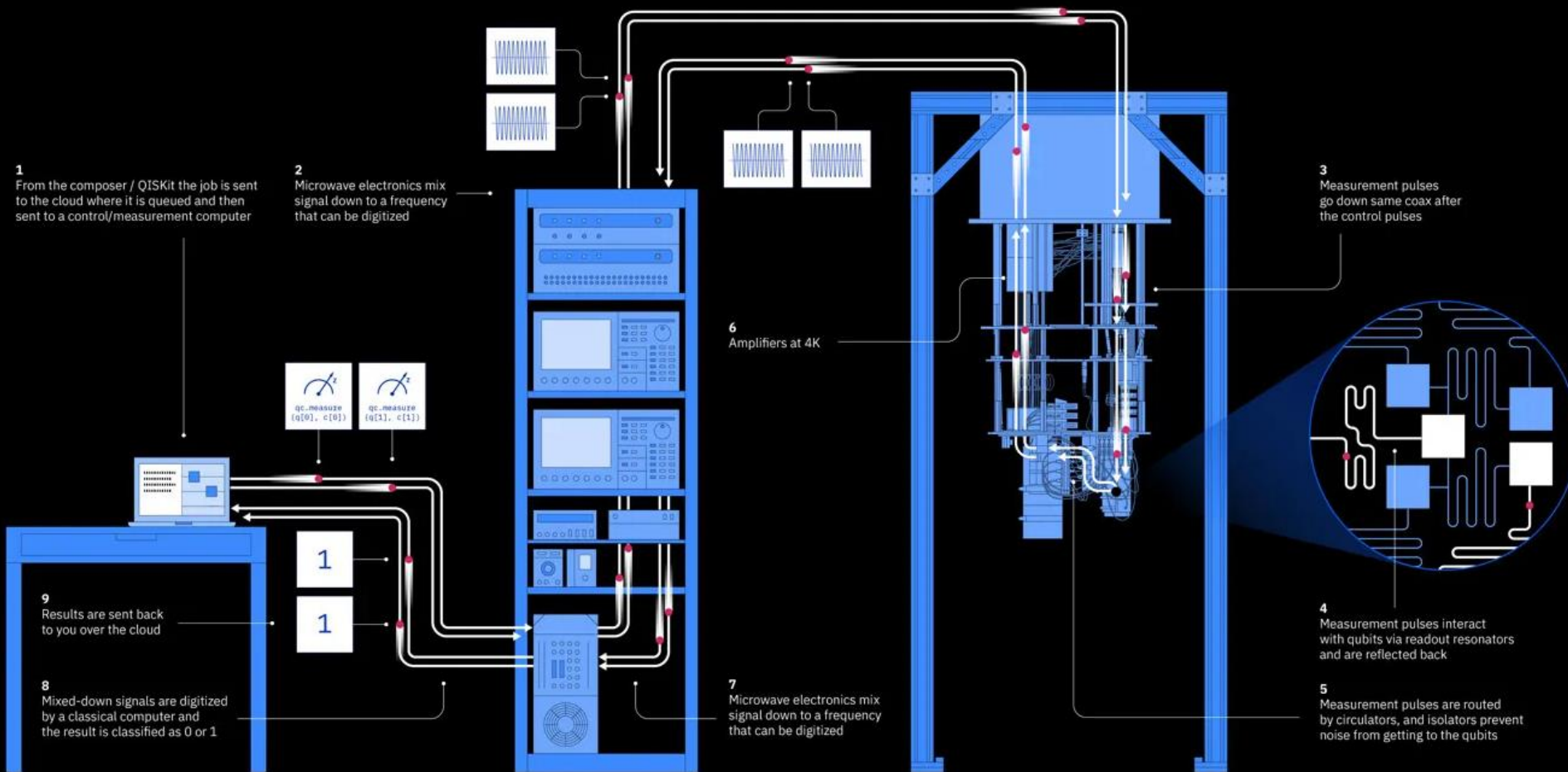
2026.05.04



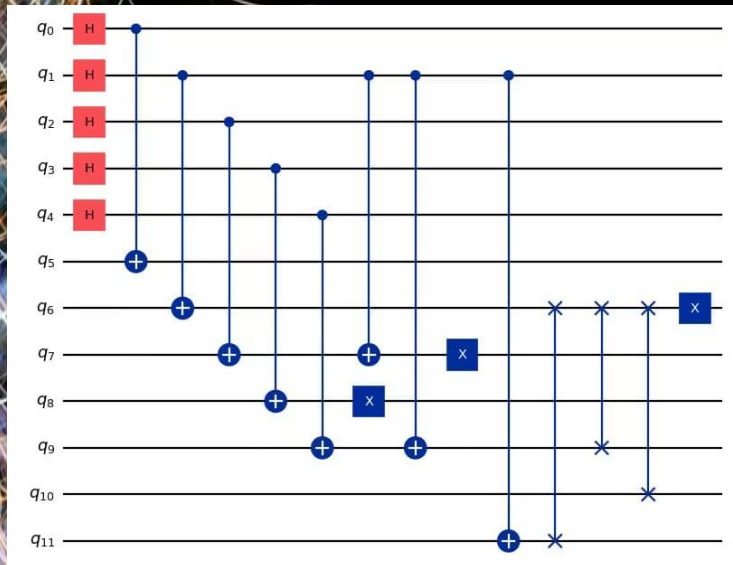
Key Information

- 超導量子計算(Superconducting Quantum Computation, SC QC)，是在極低溫下，以超導(Superconductor)製作的約瑟夫森節(Josephson Junction)為基礎，設計出非線性的LC電路作為量子位元 (Qubit)，**搭配電路設計做出可控制、可讀出、可計算的量子處理器(QPU)**。整機運作結合微波工程、低溫物理、材料、電路設計與量子控制。
- 04/20：超導量子計算是什麼？
- 04/27：超導量子位元的特性、超導量子電路
- **05/04：如何控制、校正超導量子位元？**
- 05/11：超導量子電腦硬體元件導論

IBM's Prototype



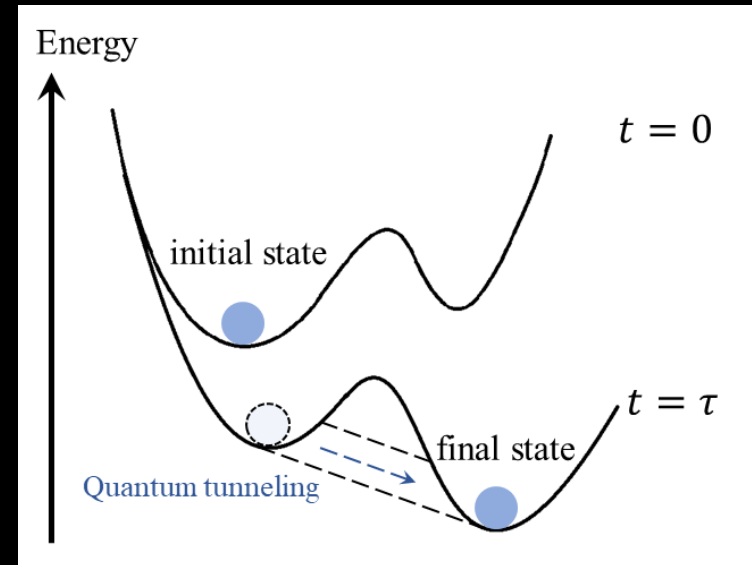
Gate Based/Circuit Model



- Algorithm: Shor, Grover,...
- Quantum Simulation
- Discrete control (logic gate)



Adiabatic/Annealing /Ising Machine



- Optimization: salesman, max-cut...
- Ground State Search
- Continuous control



Quantum Time Evolution

Physics

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi$$

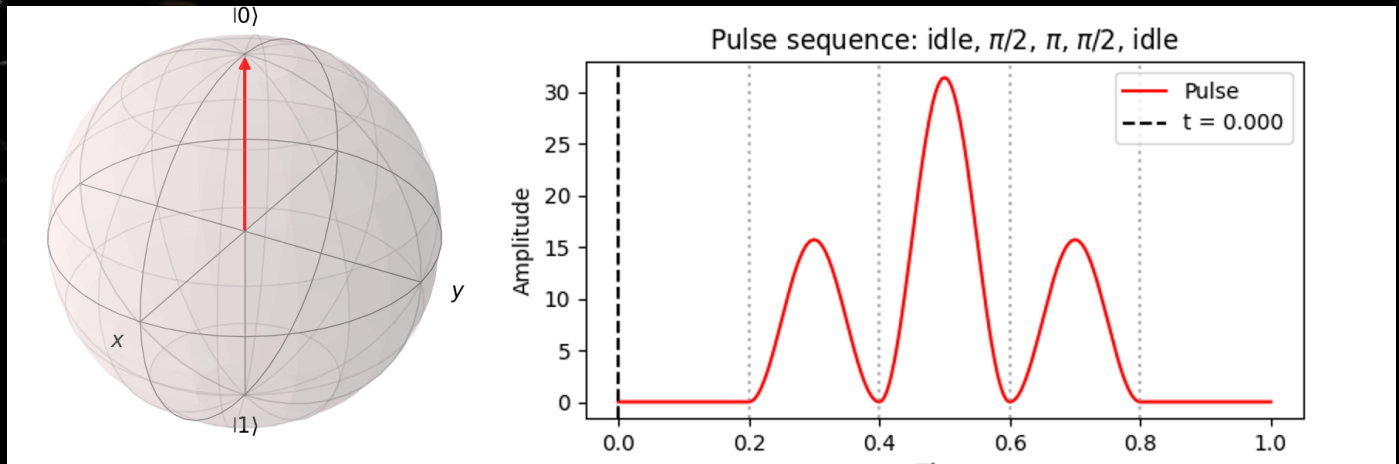
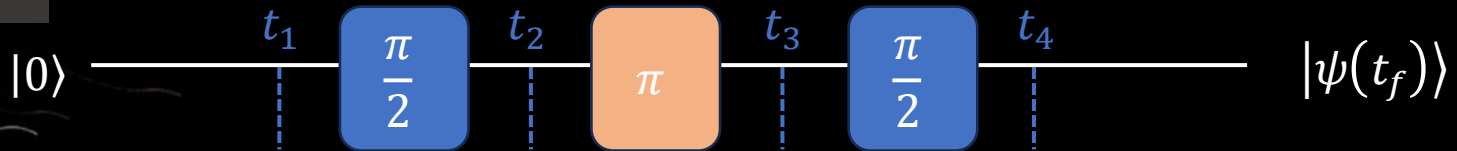
$$\psi(t_f) = \left[\exp \left(-\frac{i}{\hbar} T \left[\int_{t_i}^{t_f} \hat{H}(s) ds \right] \right) \right] \psi(t_i)$$

$$U(t_f, t_i)$$

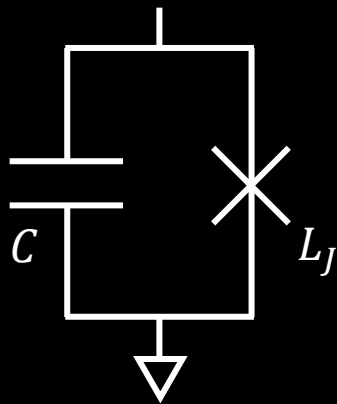
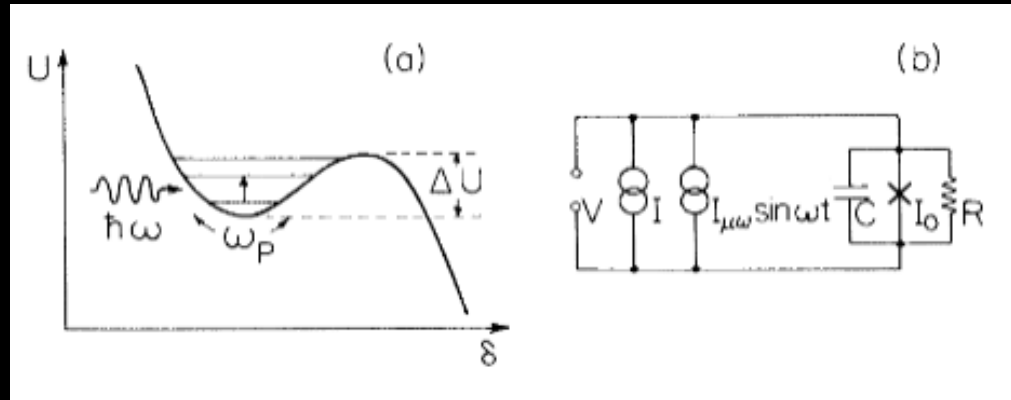
$$= U(t_f, t_4) * U(t_4, t_3) * U(t_3, t_2) * U(t_2, t_1) * U(t_1, t_i) * \psi(t_i)$$



Algorithm



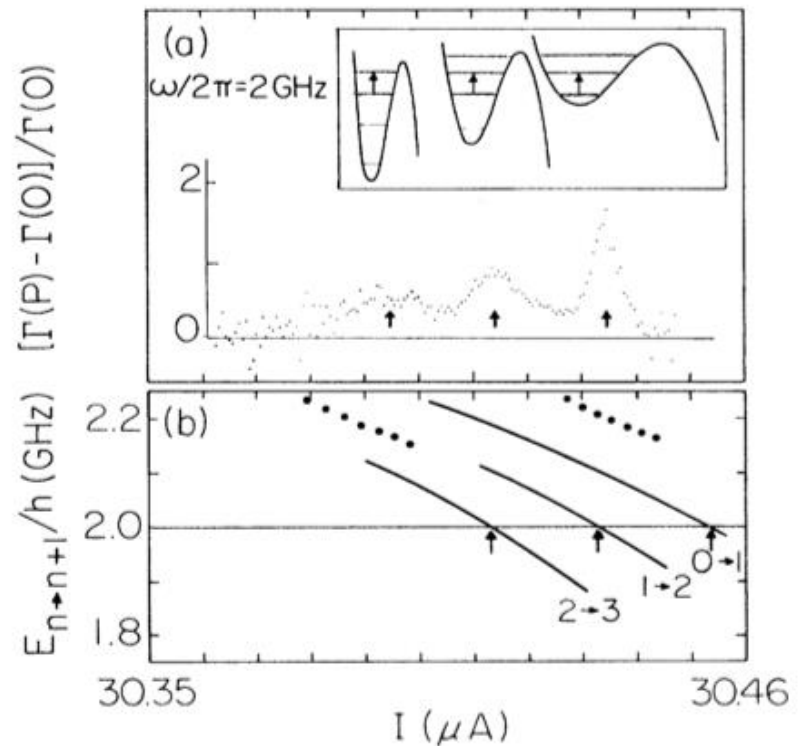
Nobel 2025



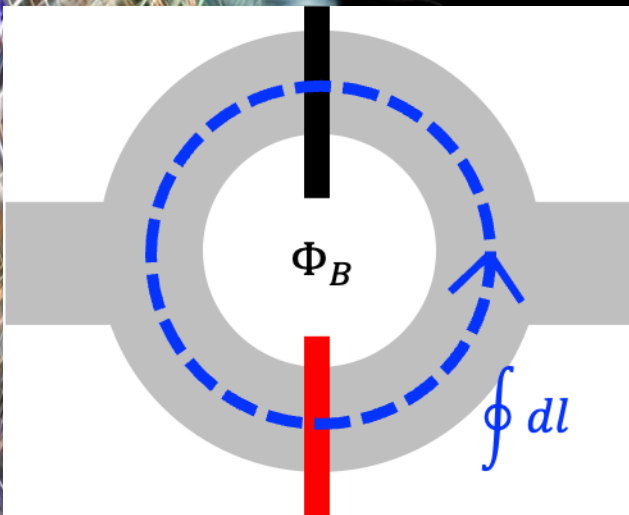
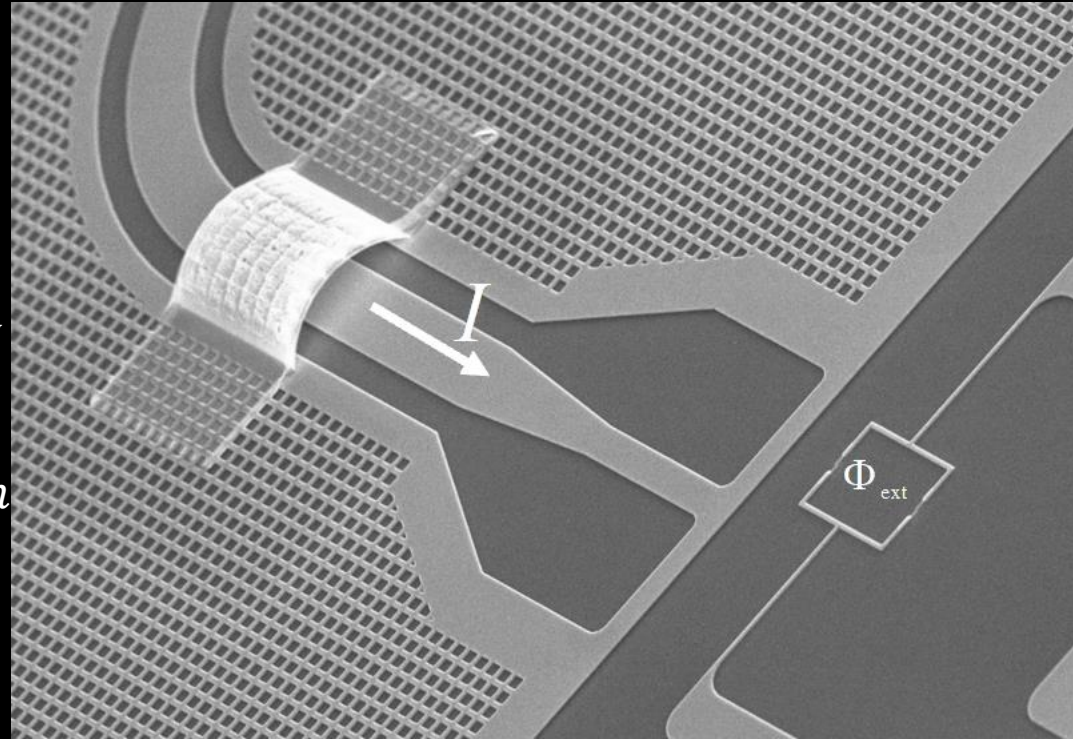
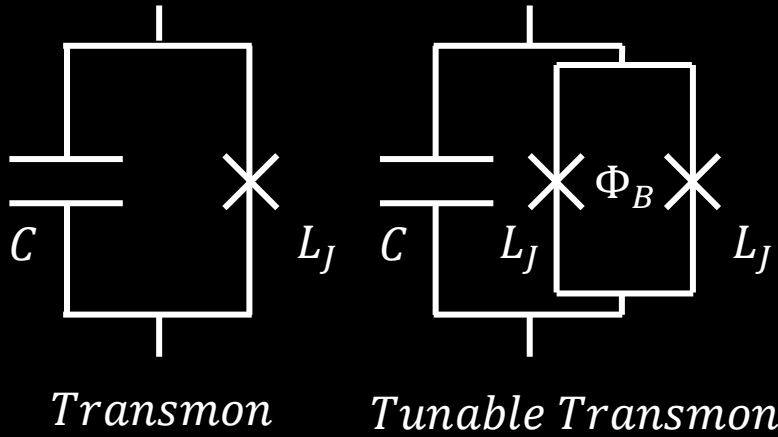
1985

$$\omega = \frac{1}{\sqrt{L_J C}} \propto \sqrt{I_c^2 - I^2}$$

10.1103/PhysRevLett.55.1543, Nobel



Josephson Junction Loop



$$\omega \propto \sqrt{\cos\left(\Delta\theta + \frac{\Phi_B}{\Phi_0}\right)}$$

$$\Phi_0 = \frac{2\pi\hbar}{2e}$$

Tunable Transmon

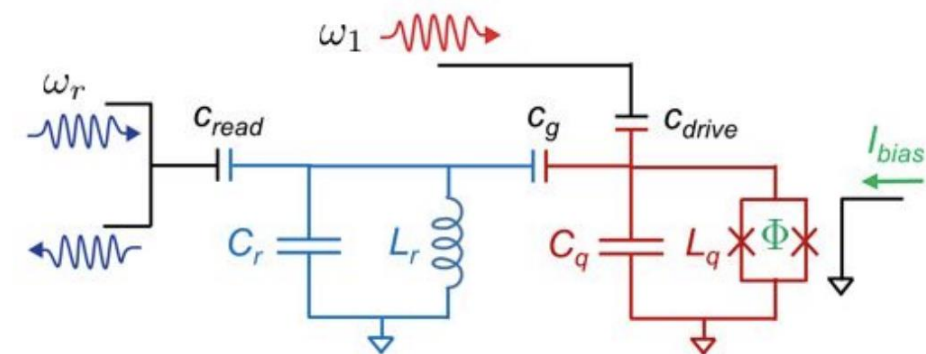
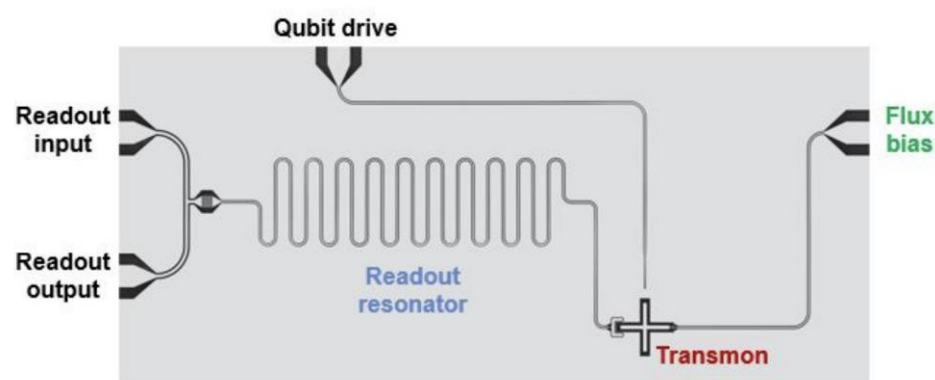
$$L_q(\Phi_B)$$

$$= \left[\frac{I_0 \pi}{2} \left| \cos \left(\frac{\Phi_B}{\Phi_0} \pi \right) \right| \right]^{-1}$$

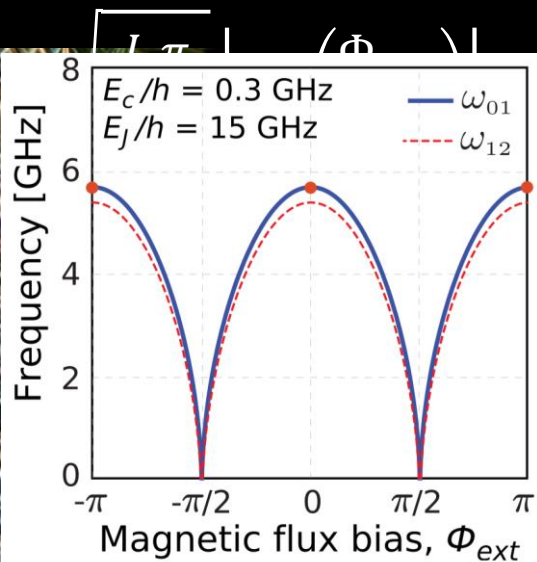
$$\mathcal{H}_q \cong \hbar \omega_q \left[a_q^\dagger a_q + \frac{1}{2} \right] - \frac{\eta}{12} * (a_q^\dagger + a_q)^4$$

$$\eta = \frac{e^2}{8C_q}$$

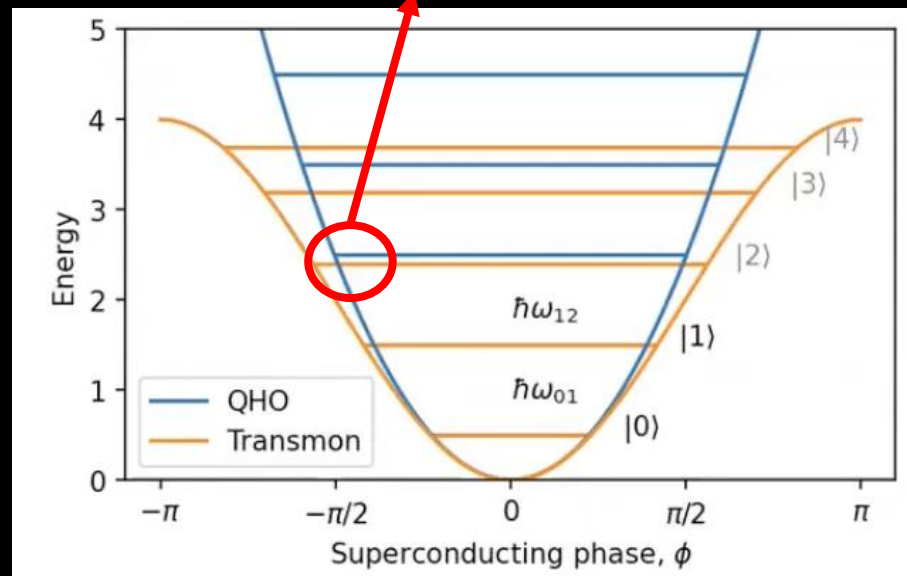
$$\omega_q(\Phi_B) = \frac{1}{\sqrt{L_q C_q}}$$



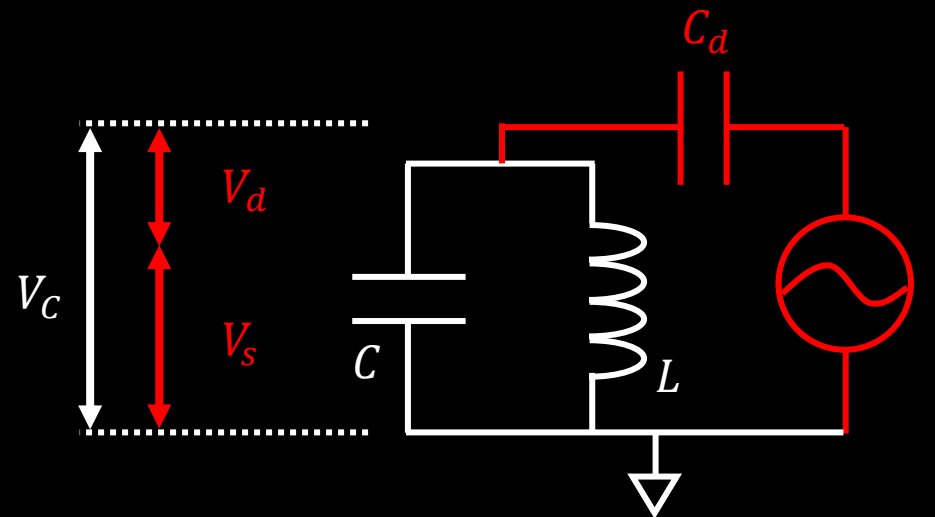
$$\Delta E^{21} - \Delta E^{10} = -\eta \sim 100 \text{ MHz}$$



10GHz



Driven



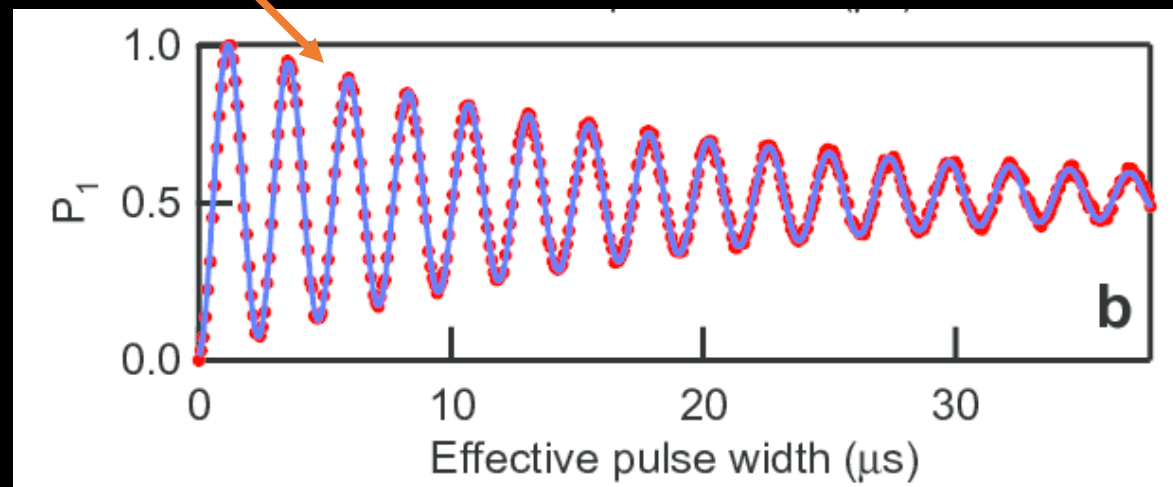
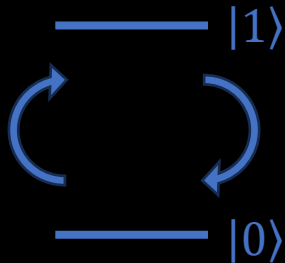
$$\mathcal{H} = \hbar\omega_\Sigma \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) + \hbar\Omega * f(t) * i(\hat{a}^\dagger - \hat{a})$$

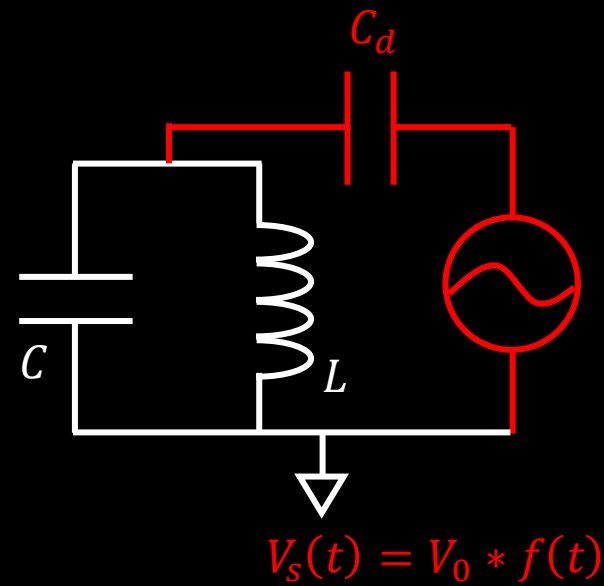
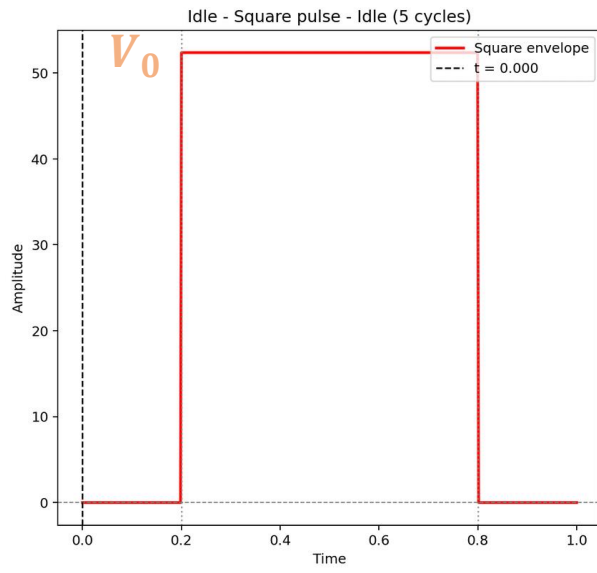
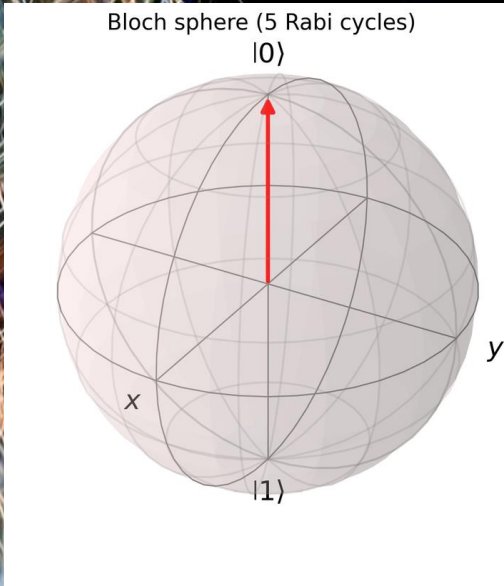
$$\text{Rabi Frequency } \Omega = C_d V_0 \sqrt{\frac{\omega_\Sigma}{2\hbar(C + C_d)}}$$

Informally, for 2-state system,

$$(\hat{a}^\dagger - \hat{a})|0\rangle = \hat{a}^\dagger|0\rangle = |1\rangle$$

$$(\hat{a}^\dagger - \hat{a})|1\rangle = -\hat{a}|1\rangle = -|0\rangle$$



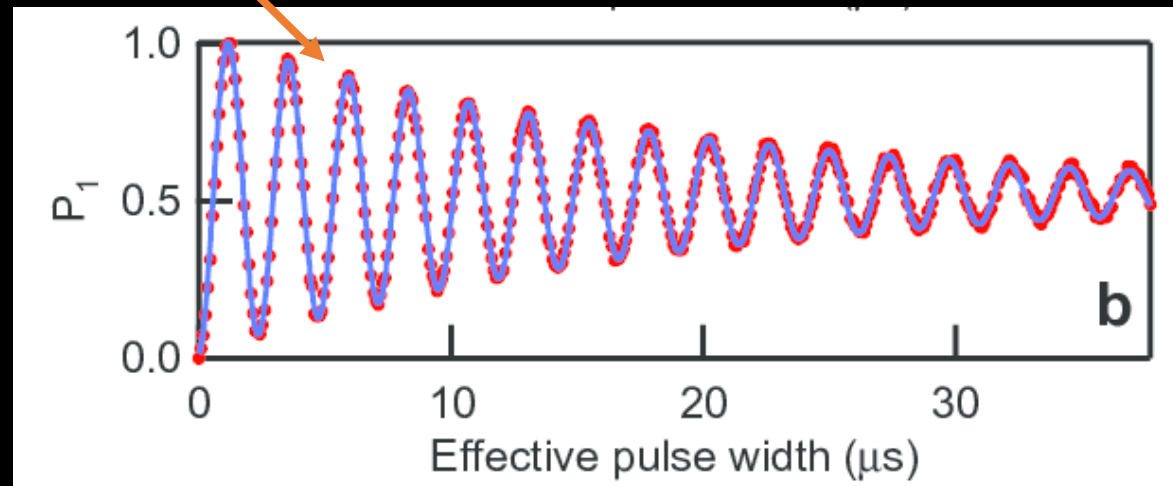
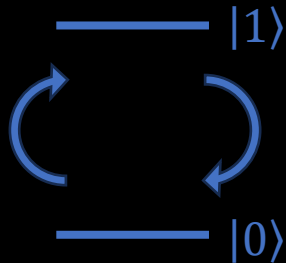


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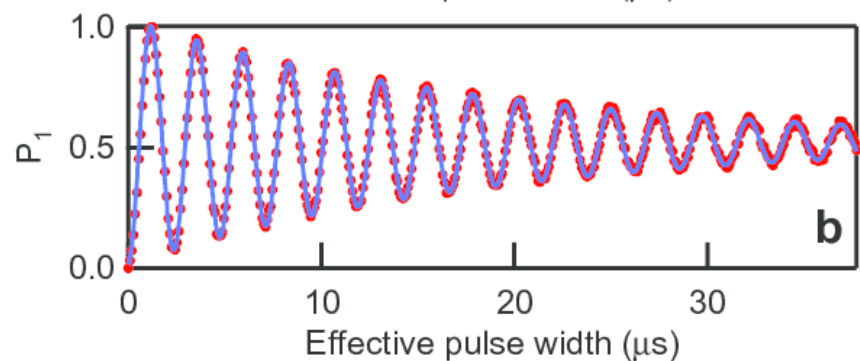
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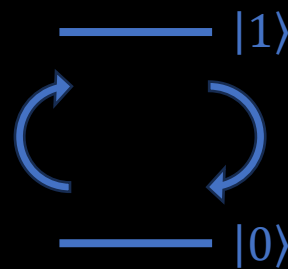
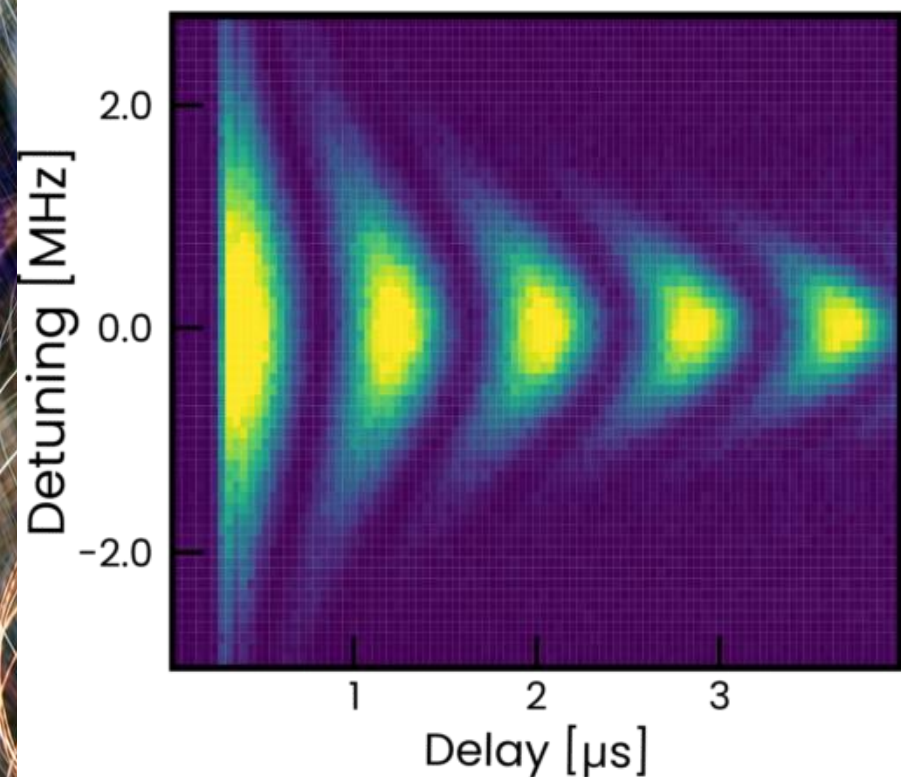
$$(\hat{a}^\dagger - \hat{a})|1\rangle = -\hat{a}|1\rangle = -|0\rangle$$



Chevron



RABI-CHEVRON

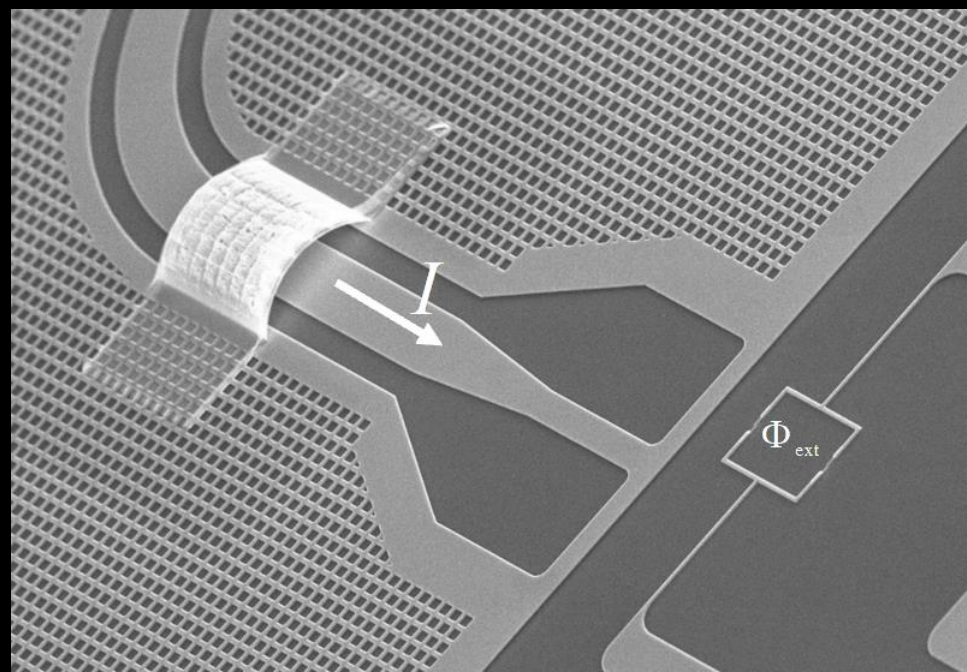


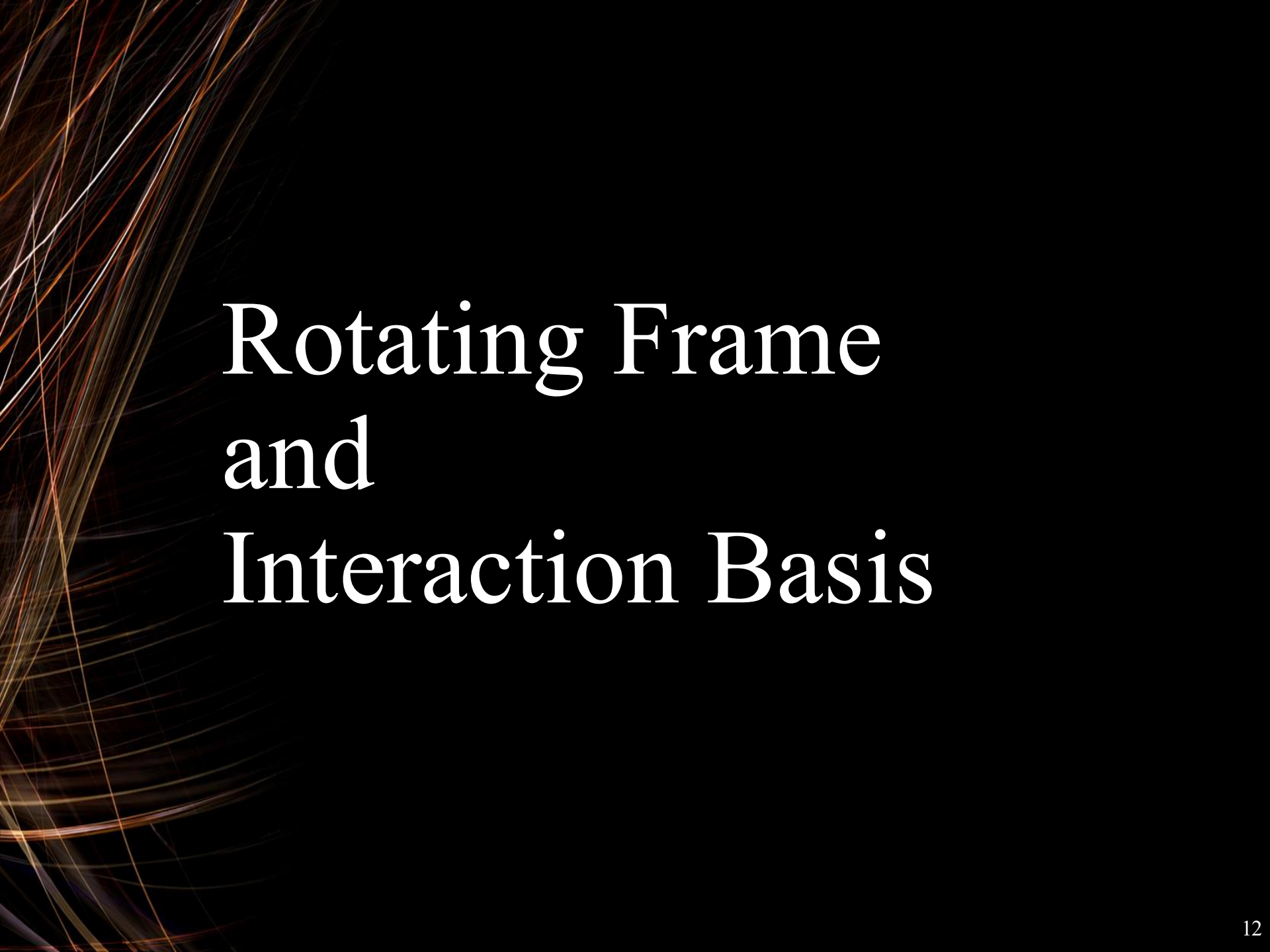
$$\omega_{\Sigma} \propto \sqrt{\cos\left(\Delta\theta + \frac{\Phi_B}{\Phi_0}\right)}$$

$$\Phi_0 = \frac{2\pi\hbar}{2e}$$

$$\mathcal{H} = \hbar\omega_{\Sigma}\left(\hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right) + \hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})$$

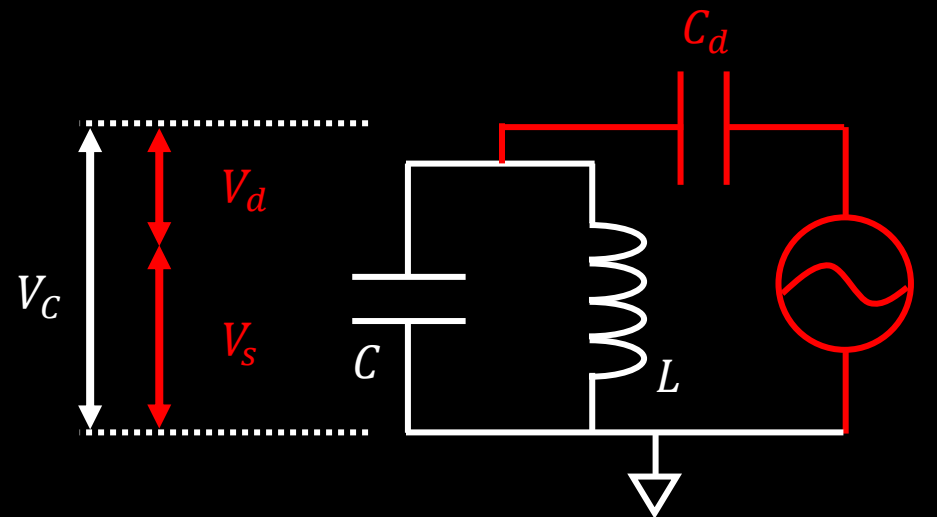
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Rotating Frame and Interaction Basis

Driven



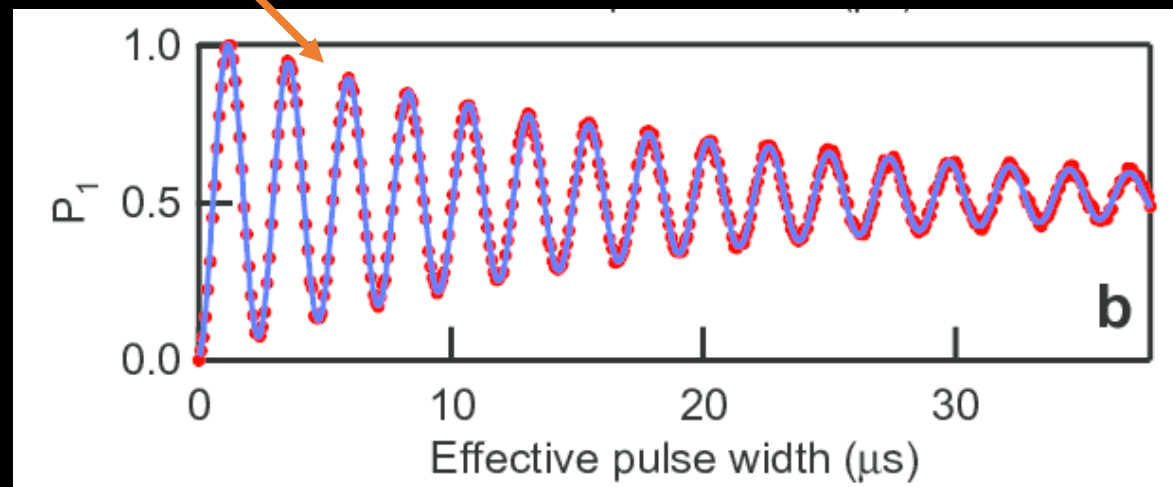
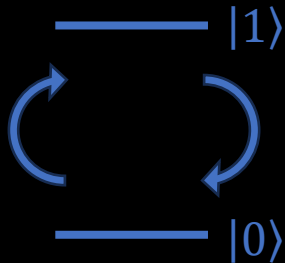
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Informally, for 2-state system,

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Quantum Phase

$$i\hbar\partial_t|\psi\rangle = \hat{H}|\psi\rangle \quad |\psi\rangle = \sum_i c^i |\psi_i\rangle \quad \hat{H}|\psi_i\rangle = E^i|\psi_i\rangle$$

$$|\psi_i(t)\rangle = \exp\left(-i\frac{E^i}{\hbar}t\right) * |\psi_i(0)\rangle$$

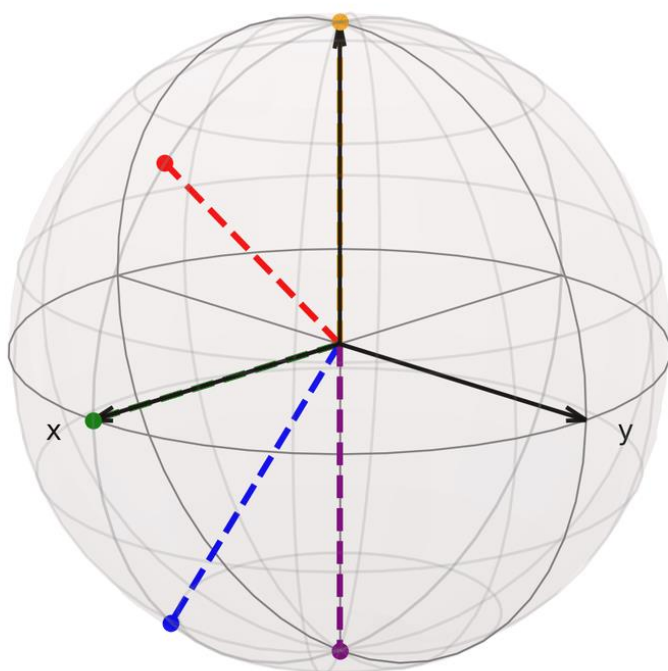
Phase

$$|\psi(t)\rangle = \left[\exp\left(-\frac{i}{\hbar}T\left[\int_0^t \hat{H}(s)ds\right]\right) \right] |\psi(0)\rangle$$

$U(t, 0)$

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \xrightarrow{\Delta t} |+\rangle = \frac{e^{-i\frac{E^0}{\hbar}t}|0\rangle + e^{-i\frac{E^1}{\hbar}t}|1\rangle}{\sqrt{2}}$$

Lab frame
z



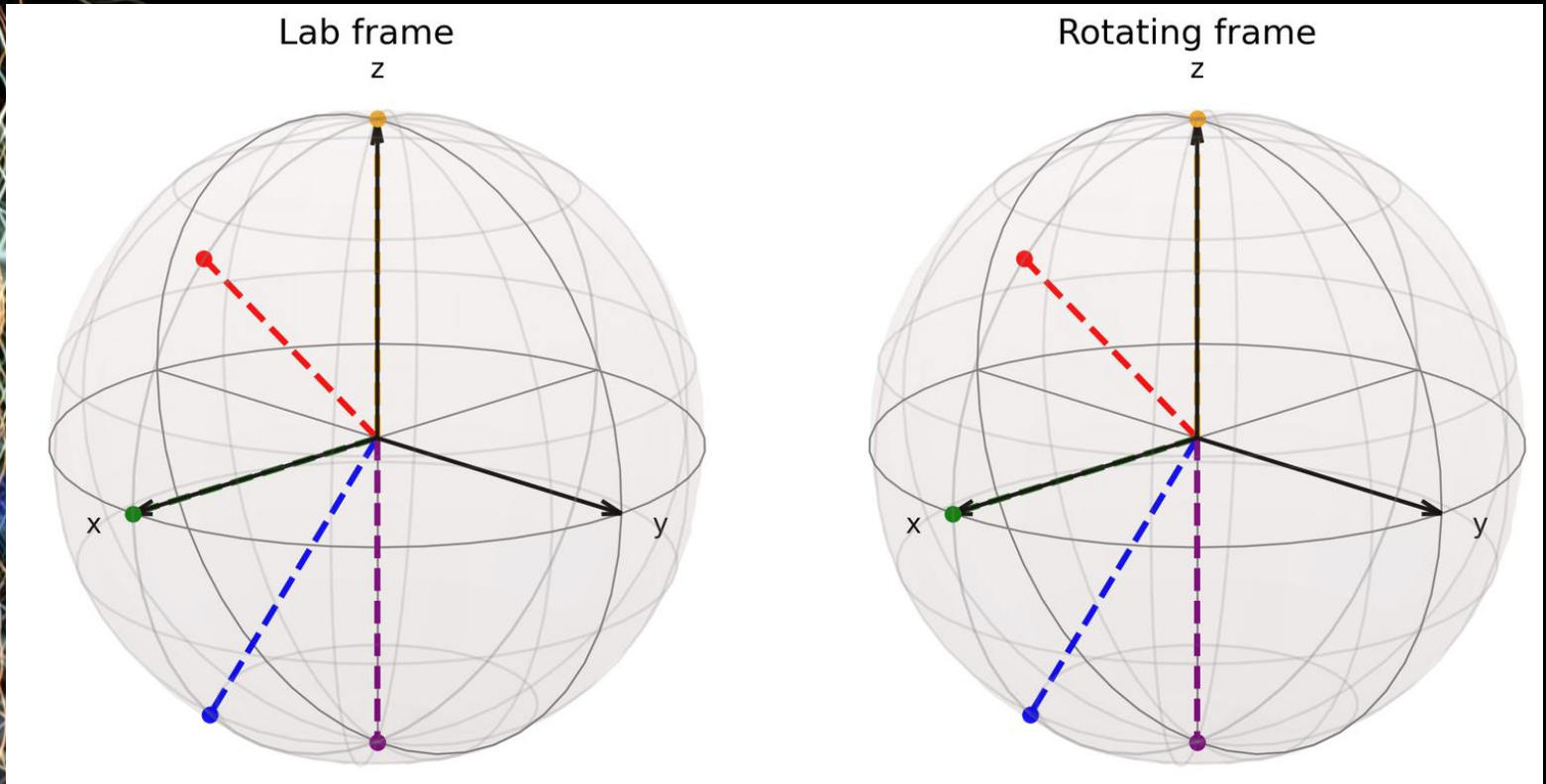
Rotating Frame

$$|\Theta(t)\rangle = |\Theta(0)\rangle$$

$$i\hbar\partial_t|\Theta\rangle = \hat{H}_r|\Theta\rangle$$

$$|\Theta(t)\rangle = \left[\exp\left(-\frac{i}{\hbar}T\left[\int_0^t \hat{H}_r(s)ds\right]\right) \right] |\Theta(0)\rangle$$

$$U_r(t, 0) = \mathbb{I} \leftrightarrow \hat{H}_r = 0$$



Rotating Frame

$$|\Theta(t)\rangle = |\Theta(0)\rangle \text{ Goal}$$

$$|\Theta(t)\rangle \equiv U(0,t)|\psi(t)\rangle = U(0,t)U(t,0)|\psi(0)\rangle = |\psi(0)\rangle = |\Theta(0)\rangle \checkmark$$

$$|\Theta(t)\rangle = U^\dagger(t,0)|\psi(t)\rangle$$

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

$$i\hbar \frac{\partial}{\partial t} [U(t,0)|\psi(0)\rangle] = \hat{H} [U(t,0)|\psi(0)\rangle]$$

$$i\hbar \frac{\partial}{\partial t} U(t,0) = \hat{H} U(t,0)$$

$$\rightarrow -i\hbar \frac{\partial}{\partial t} U^\dagger(t,0) = U^\dagger(t,0) \hat{H}$$

$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = i\hbar \frac{\partial}{\partial t} [U^\dagger(t,0)|\psi(t)\rangle]$$

$$= \left[i\hbar \frac{\partial}{\partial t} U^\dagger(t,0) \right] |\psi(t)\rangle + U^\dagger(t,0) * i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle$$

$$= -U^\dagger(t,0) \hat{H} |\psi(t)\rangle + U^\dagger(t,0) * \hat{H} |\psi(t)\rangle = 0$$

$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = 0$$

$$|\psi(t)\rangle = U(t,0)|\psi(0)\rangle$$

$$U^\dagger(t,0)U(0,t) = \mathbb{I}$$

$$U^\dagger(t,0) = U(0,t)$$

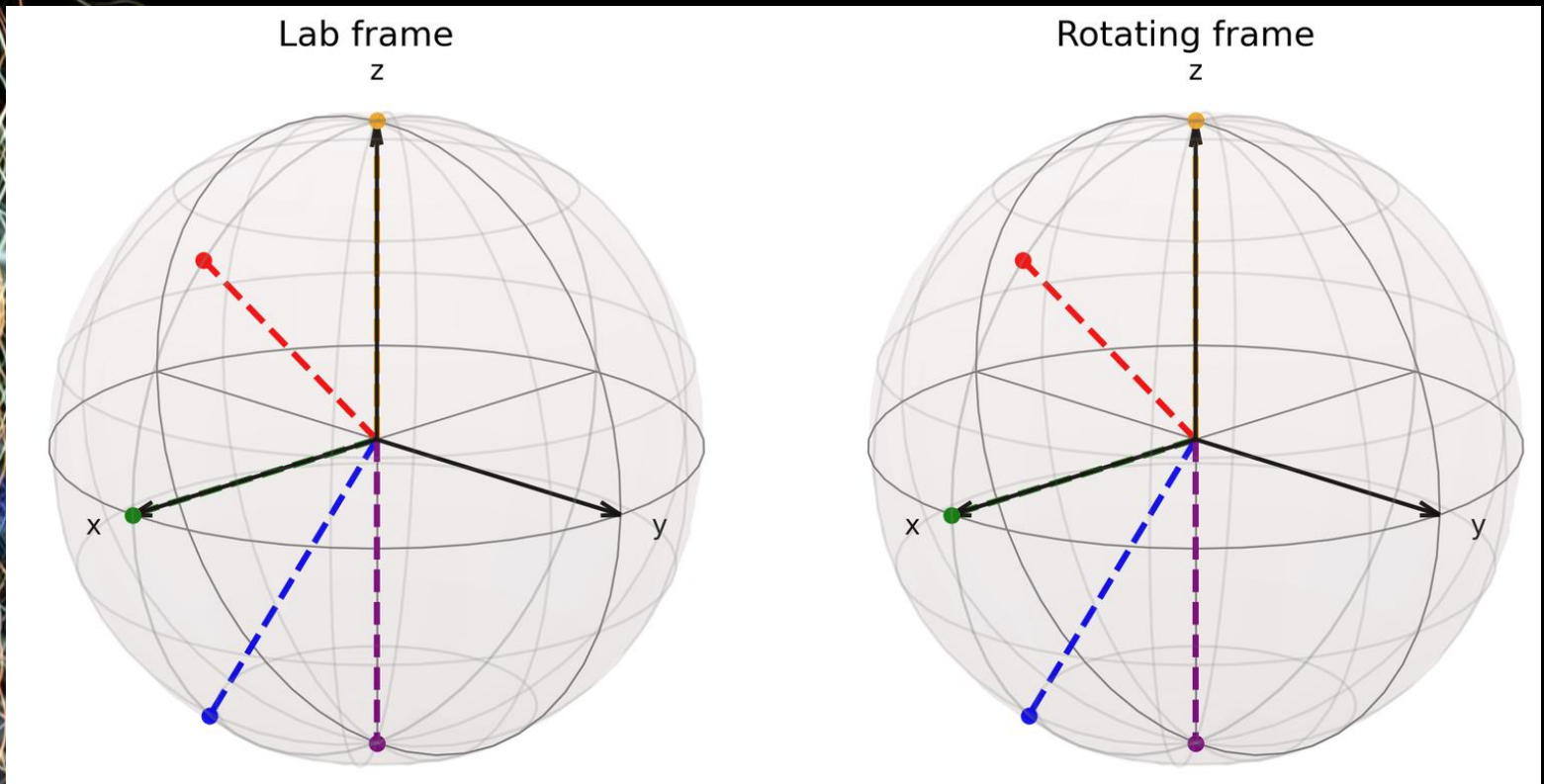
Rotating Frame

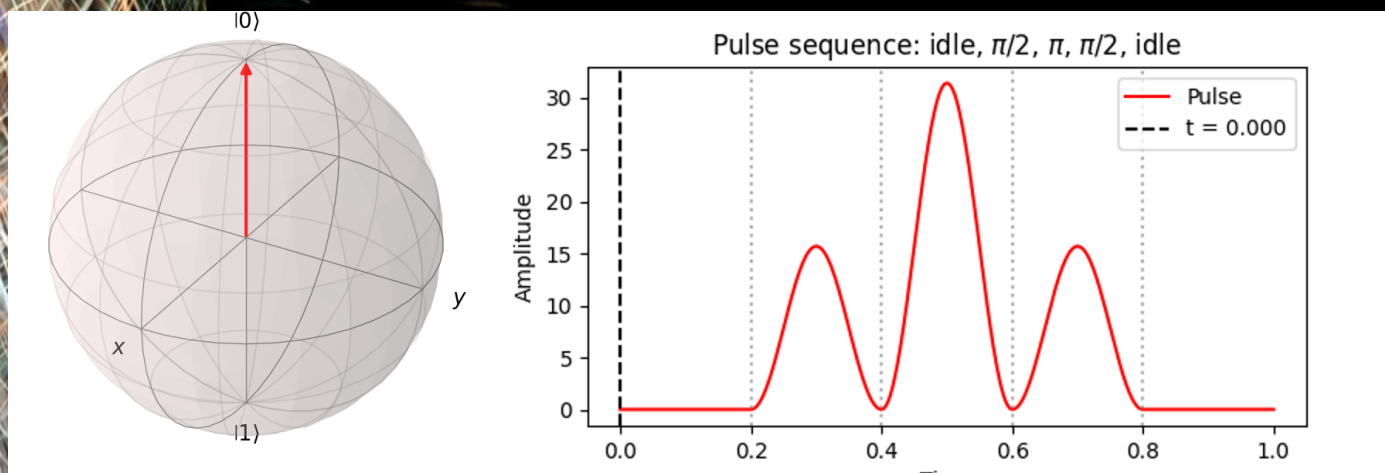
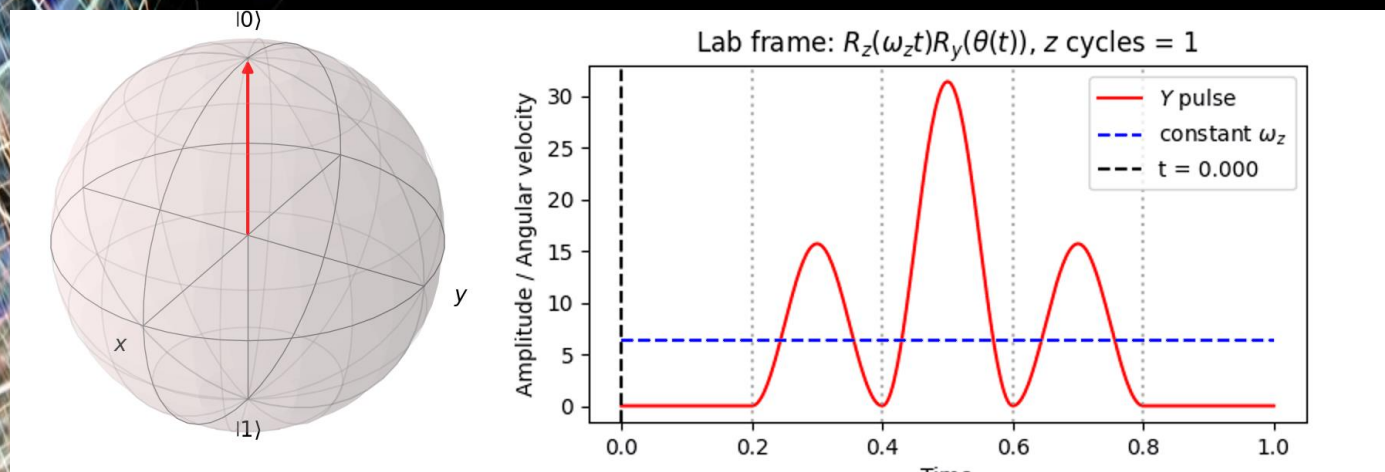
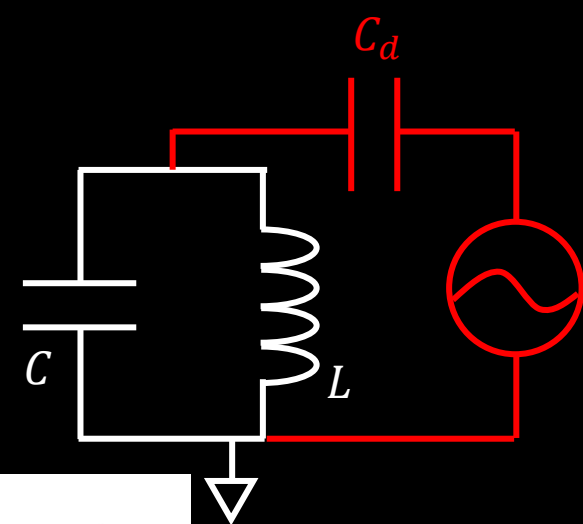
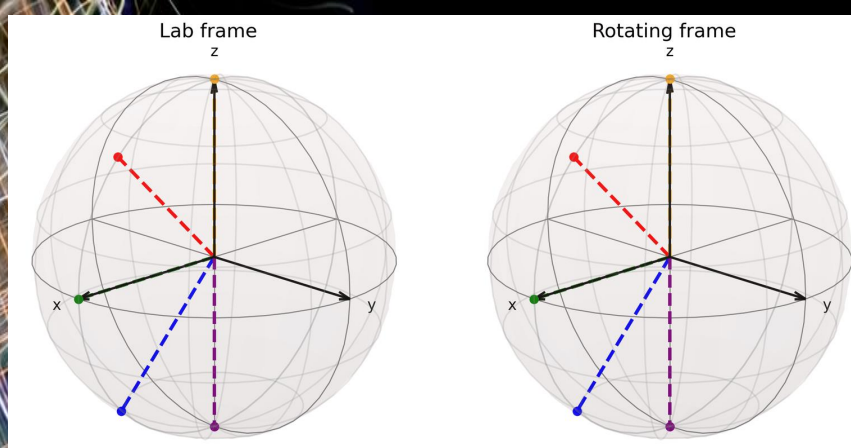
$$|\Theta(t)\rangle = U^\dagger(t, 0)|\psi(t)\rangle$$

$$i\hbar\partial_t|\Theta\rangle = \hat{H}_r|\Theta\rangle = \mathbf{0}$$

$$|\Theta(t)\rangle = \left[\exp\left(-\frac{i}{\hbar} T \left[\int_0^t \hat{H}_r(s) ds \right] \right) \right] |\Theta(0)\rangle$$

$$U_r(t, 0) = \mathbb{I} \leftrightarrow \hat{H}_r = 0$$





Interaction Basis

$$\hat{H} = \underbrace{\hbar\omega_\Sigma \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right)}_{\hat{H}_0} + \underbrace{\hbar\Omega * f(t) * i(\hat{a}^\dagger - \hat{a})}_{\hat{V}(t)}$$

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle = [\hat{H}_0 + \hat{V}(t)] |\psi(t)\rangle$$

$$|\Theta(t)\rangle \equiv U_0^\dagger(t, 0) |\psi(t)\rangle$$

$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = i\hbar \frac{\partial}{\partial t} [U_0^\dagger(t, 0) |\psi(t)\rangle]$$

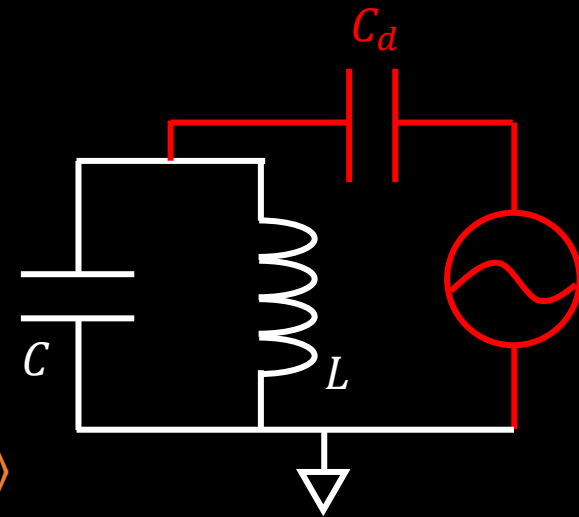
$$= \left[i\hbar \frac{\partial}{\partial t} U_0^\dagger(t, 0) \right] |\psi(t)\rangle + U_0^\dagger(t, 0) * i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle$$

$$= -U_0^\dagger(t, 0) \hat{H}_0 |\psi(t)\rangle + U_0^\dagger(t, 0) * [\hat{H}_0 + \hat{V}(t)] |\psi(t)\rangle$$

$$= U_0^\dagger(t, 0) * \hat{V}(t) |\psi(t)\rangle$$

$$= \underbrace{U_0^\dagger(t, 0) * \hat{V}(t) * U_0(t, 0)}_{\hat{H}_r} * \underbrace{U_0^\dagger(t, 0) |\psi(t)\rangle}_{|\Theta(t)\rangle}$$

$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = \hat{H}_r |\Theta(t)\rangle$$



$$i\hbar \frac{\partial}{\partial t} U_0(t, 0) = \hat{H}_0 U_0(t, 0)$$

$$\rightarrow -i\hbar \frac{\partial}{\partial t} U_0^\dagger(t, 0) = U_0^\dagger(t, 0) \hat{H}$$

Interaction Basis

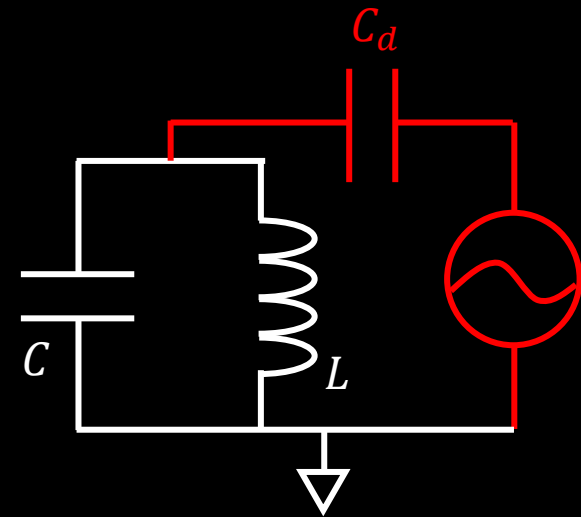
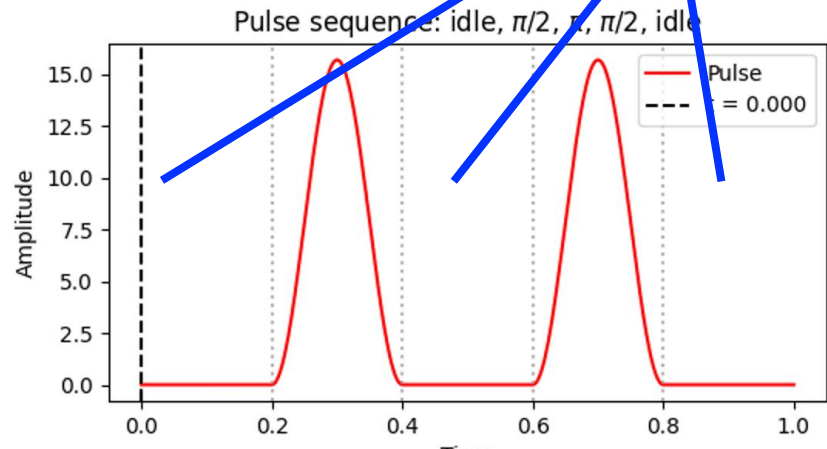
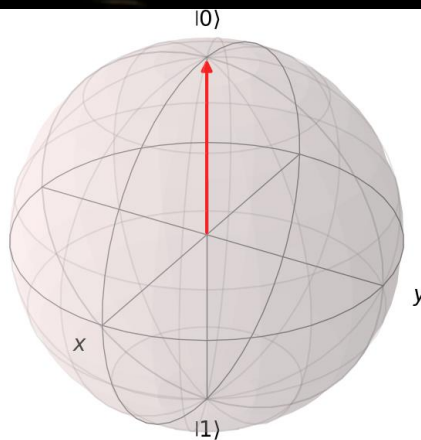
$$\hat{H} = \underbrace{\hbar\omega_{\Sigma}\left(\hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right)}_{\hat{H}_0} + \underbrace{\hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})}_{\hat{V}(t)}$$

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = [\hat{H}_0 + \hat{V}(t)] |\psi(t)\rangle$$

$$|\Theta(t)\rangle \equiv U_0^{\dagger}(t, 0) |\psi(t)\rangle$$

$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = \hat{H}_r |\Theta(t)\rangle$$

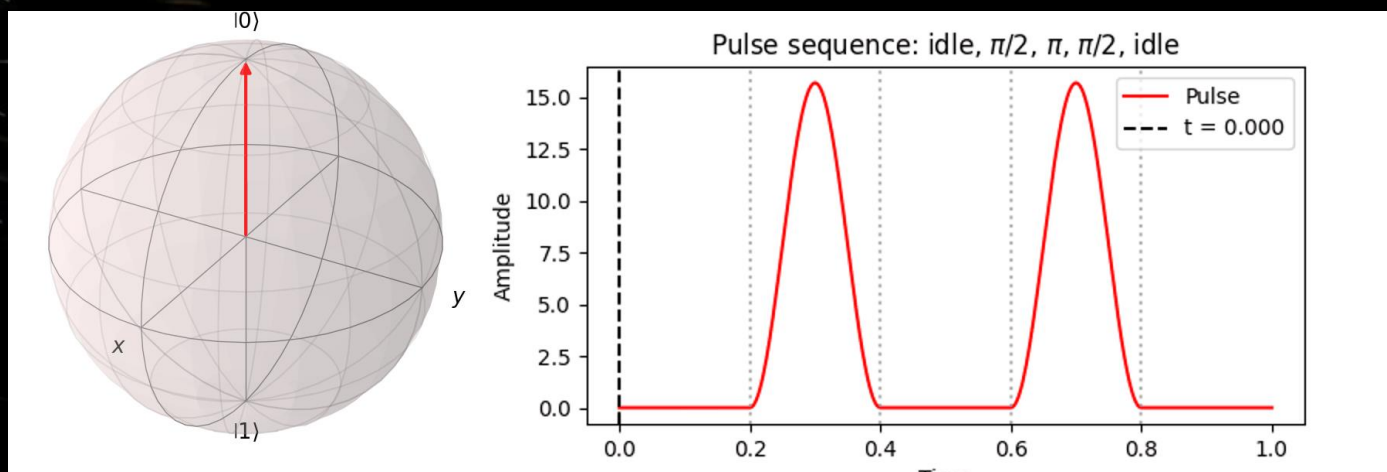
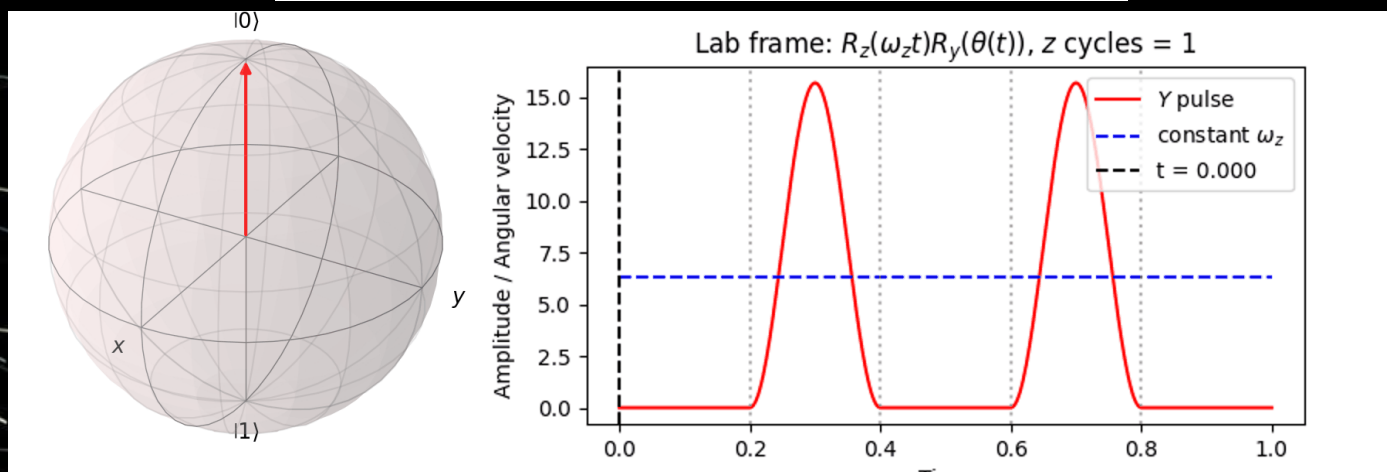
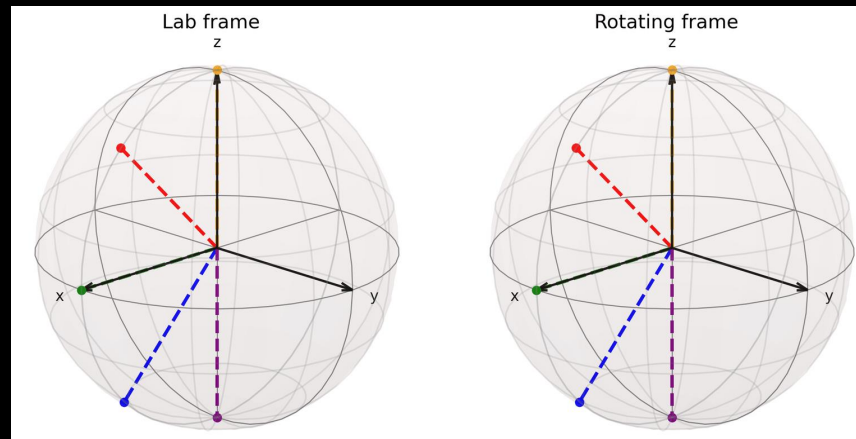
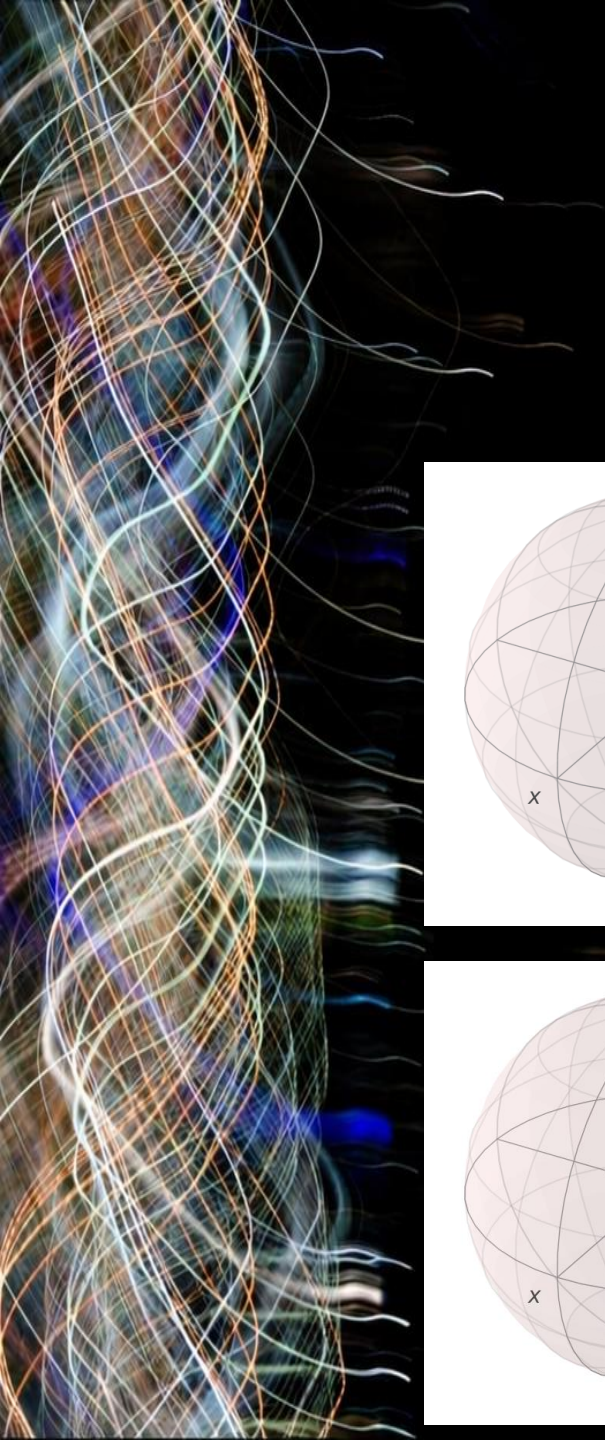
$$|\Theta(t)\rangle = \left[\exp\left(-\frac{i}{\hbar} T \left[\int_0^t \hat{H}_r(s) ds \right] \right) \right] |\Theta(0)\rangle$$

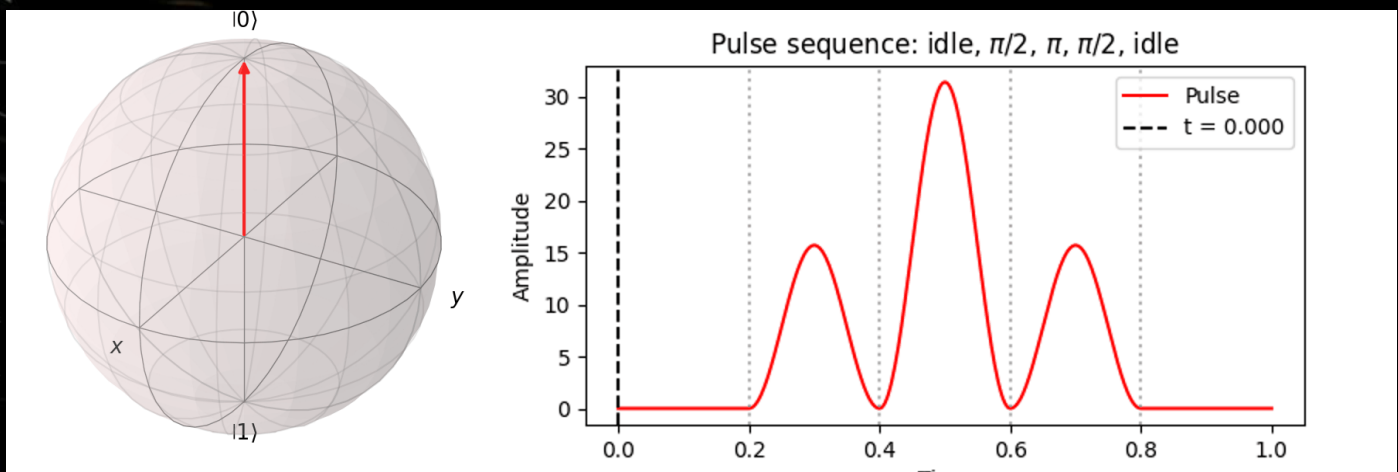
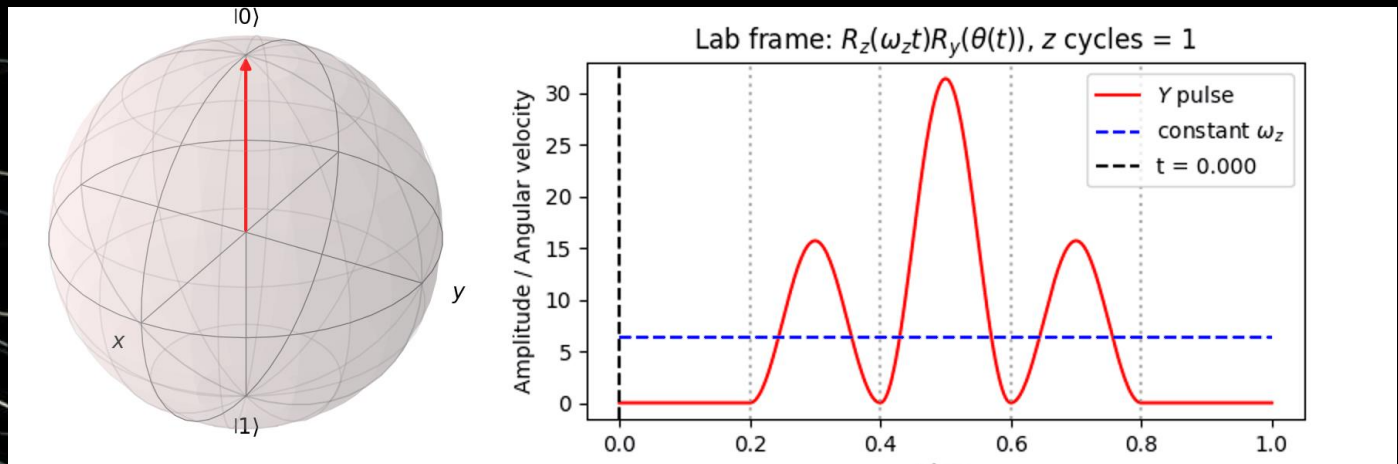
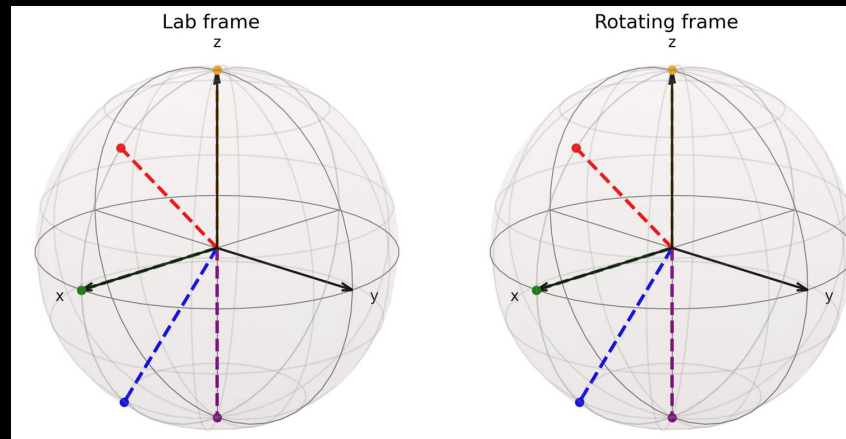
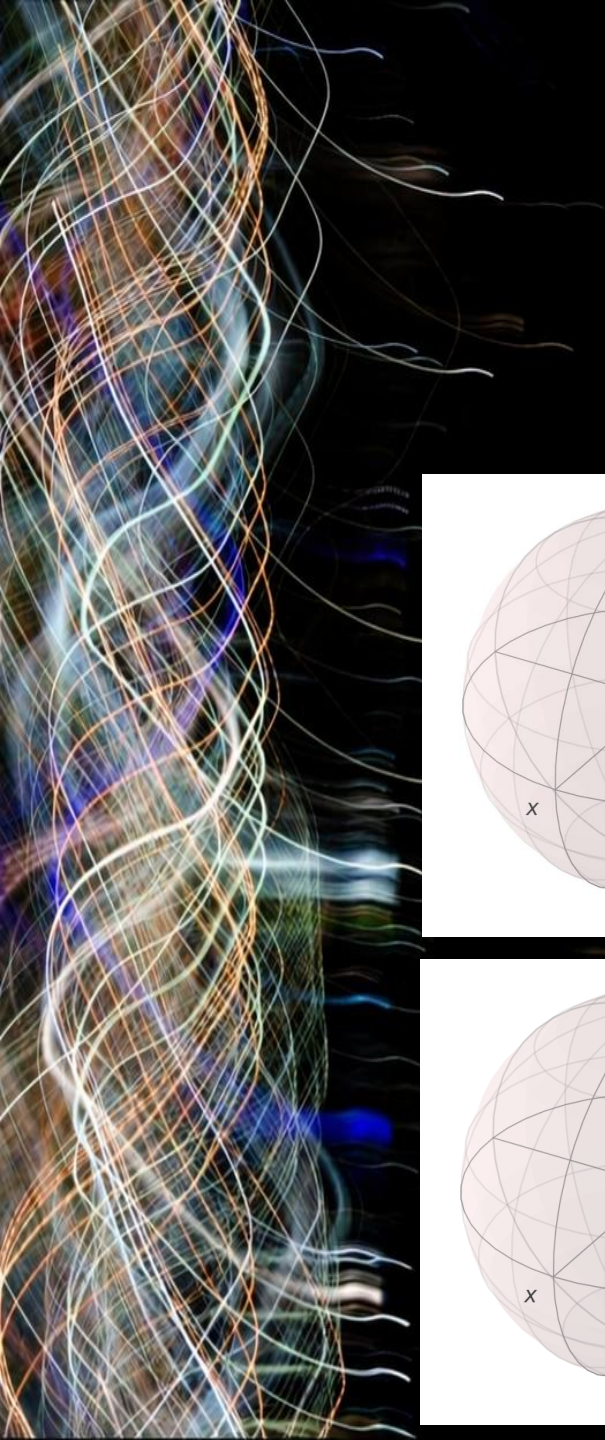


$$i\hbar \frac{\partial}{\partial t} U_0(t, 0) = \hat{H}_0 U_0(t, 0)$$

$$\hat{H}_r = U_0^{\dagger} * \hat{V}(t) * U_0$$

$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = 0$$



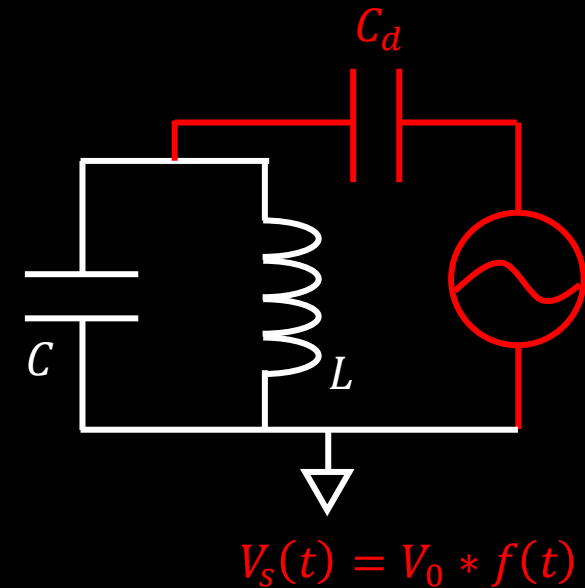


Driving Frequency

$$\hat{H} = \hbar\omega_{\Sigma} \left(\hat{a}^{\dagger} \hat{a} + \frac{1}{2} \right) + \hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})$$

$$\hat{H}_r = U_0^{\dagger} * [\hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})] * U_0$$

$$= \hbar\Omega * f(t) * i(e^{i\omega_{\Sigma}} \hat{a}^{\dagger} - e^{-i\omega_{\Sigma}} \hat{a})$$

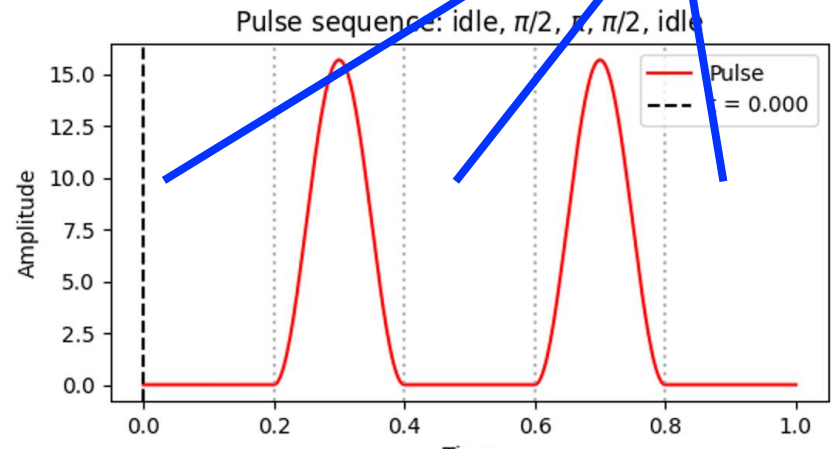
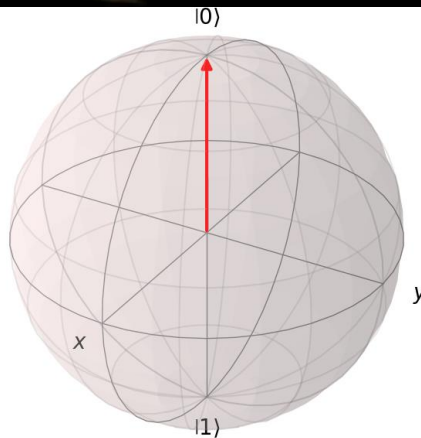


$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = \hat{H}_r |\Theta(t)\rangle$$

$$\hat{H}_r = U_0^{\dagger} * \hat{V}(t) * U_0$$

$$|\Theta(t)\rangle = \left[\exp \left(-\frac{i}{\hbar} T \left[\int_0^t \hat{H}_r(s) ds \right] \right) \right] |\Theta(0)\rangle$$

$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = 0$$

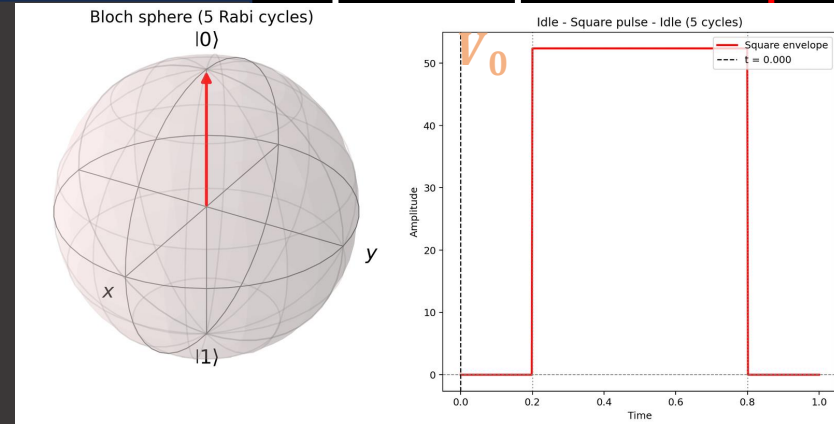


Driving Frequency

$$\hat{H} = \hbar\omega_{\Sigma} \left(\hat{a}^{\dagger} \hat{a} + \frac{1}{2} \right) + \hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})$$

$$\hat{H}_r = U_0^{\dagger} * [\hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})] * U_0$$

$$= \hbar\Omega * f(t) * i(e^{i\omega_{\Sigma}} \hat{a}^{\dagger} - e^{-i\omega_{\Sigma}} \hat{a})$$

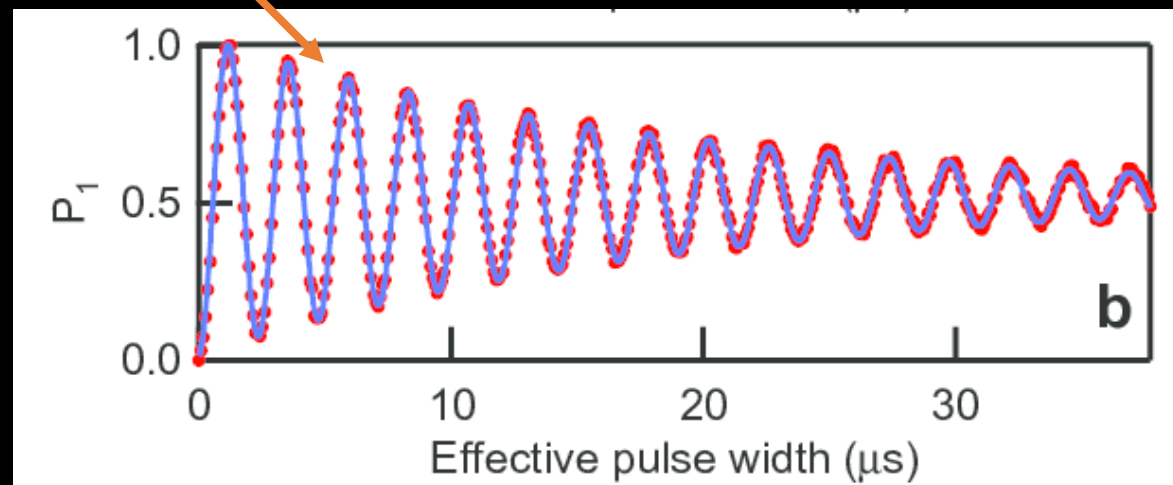
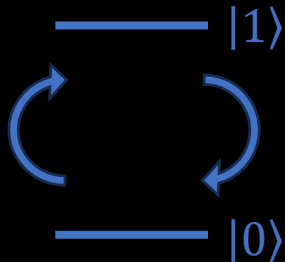


$$\text{Rabi Frequency } \Omega = C_d V_0 \sqrt{\frac{\omega_{\Sigma}}{2\hbar(C + C_d)}}$$

Informally, for 2-state system,

$$(\hat{a}^{\dagger} - \hat{a})|0\rangle = \hat{a}^{\dagger}|0\rangle = |1\rangle$$

$$(\hat{a}^{\dagger} - \hat{a})|1\rangle = -\hat{a}|1\rangle = -|0\rangle$$



Driving Frequency

$$\hat{H} = \hbar\omega_{\Sigma} \left(\hat{a}^{\dagger} \hat{a} + \frac{1}{2} \right) + \hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})$$

$$\hat{H}_r = U_0^{\dagger} * [\hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})] * U_0$$

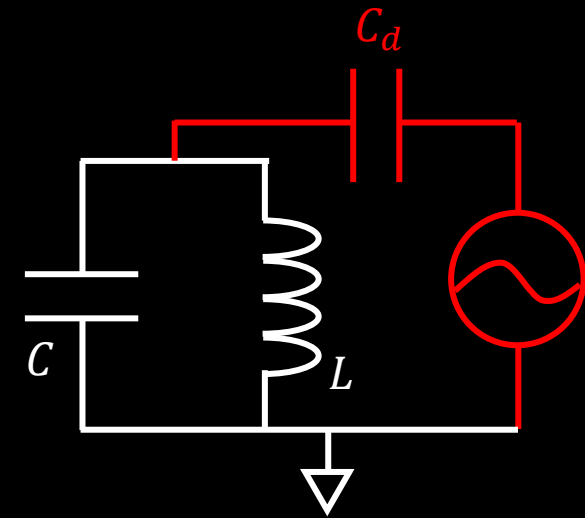
$$= \hbar\Omega * f(t) * i(e^{i\omega_{\Sigma}t} \hat{a}^{\dagger} - e^{-i\omega_{\Sigma}t} \hat{a})$$

$$= \hbar\Omega * f(t) * i[(\cos \omega_{\Sigma}t + i \sin \omega_{\Sigma}t) \hat{a}^{\dagger} - c.c.]$$

$$f(t) = \sin(\omega_{\Sigma}t)$$

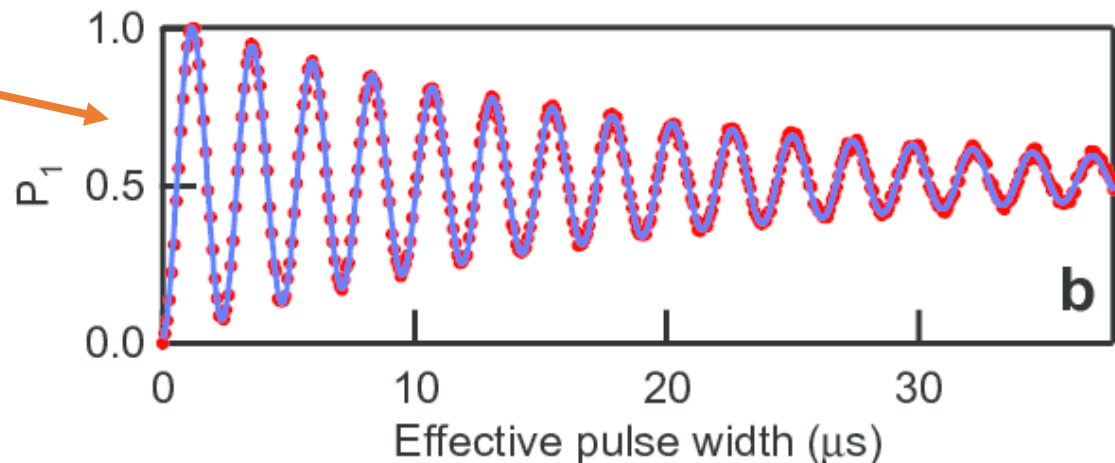
$$= \hbar\Omega * [\cos(0) \hat{a}^{\dagger} + \cos(2\omega_{\Sigma}t) \hat{a}^{\dagger} + \dots - c.c.]$$

$$= \hbar\Omega * (\hat{a}^{\dagger} - \hat{a}) + \cos(2\omega_{\Sigma}t) [...]$$



$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

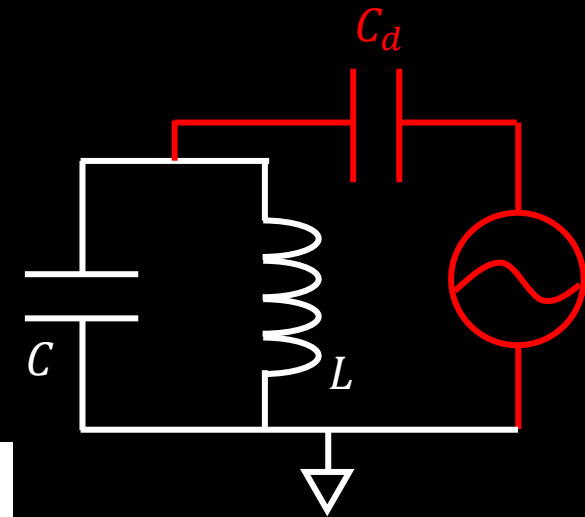
$$\sin A \cos B = \frac{1}{2} [\sin(A - B) - \sin(A + B)]$$



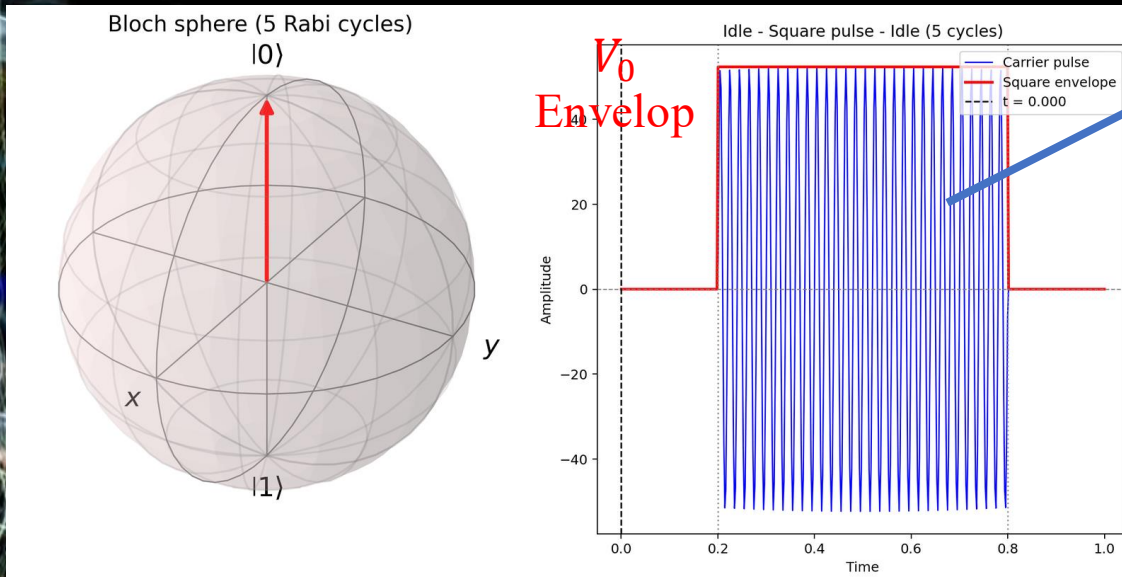
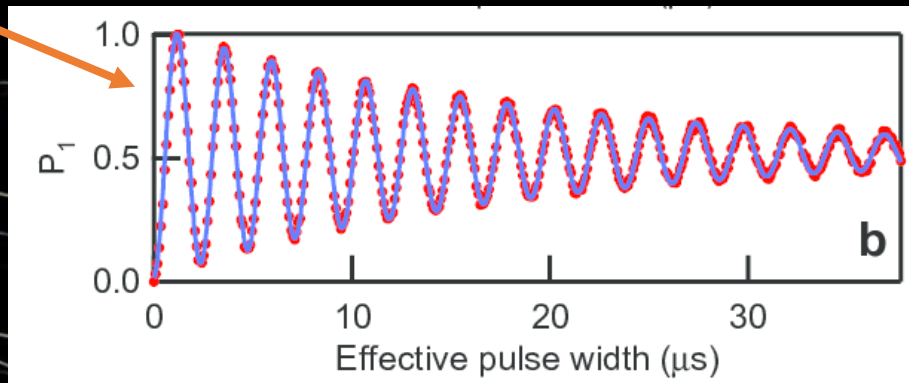
Driving Frequency

$$\hat{H} = \hbar\omega_{\Sigma} \left(\hat{a}^{\dagger} \hat{a} + \frac{1}{2} \right) + \hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})$$

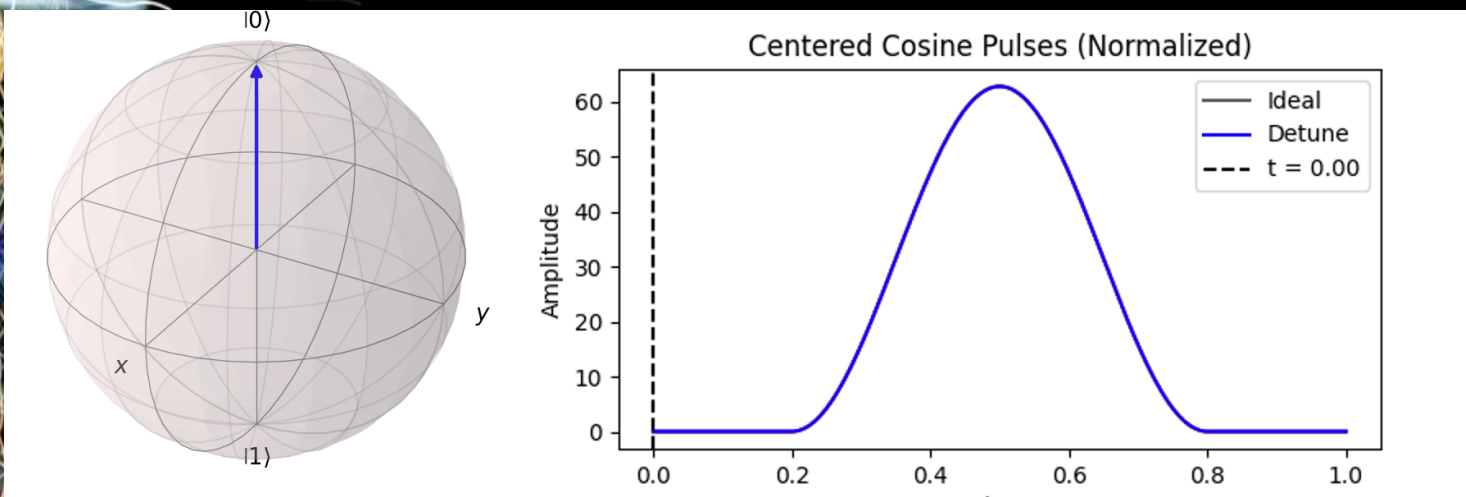
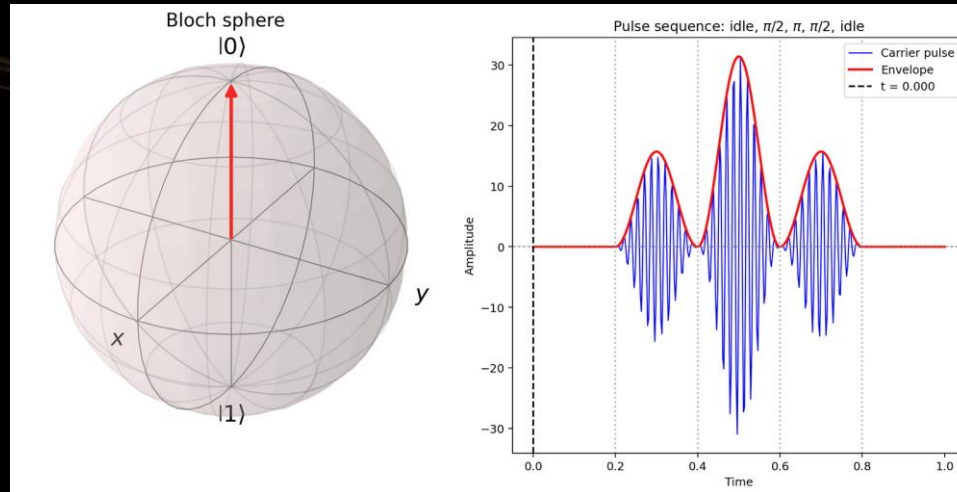
$$\hat{H}_r = \hbar\Omega * (\hat{a}^{\dagger} - \hat{a}) + \dots$$



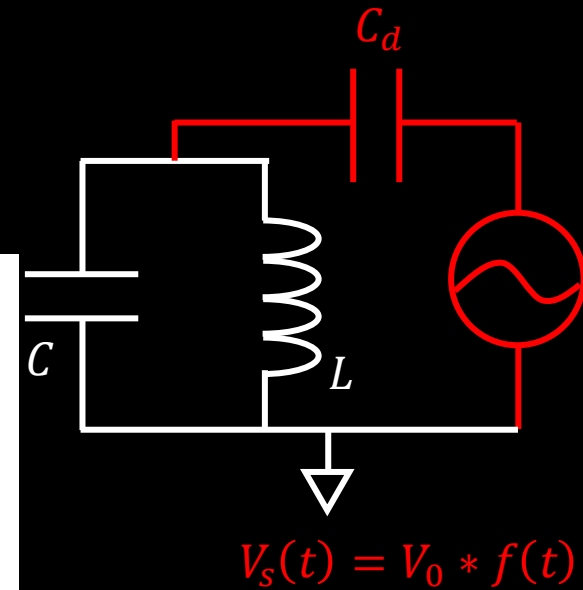
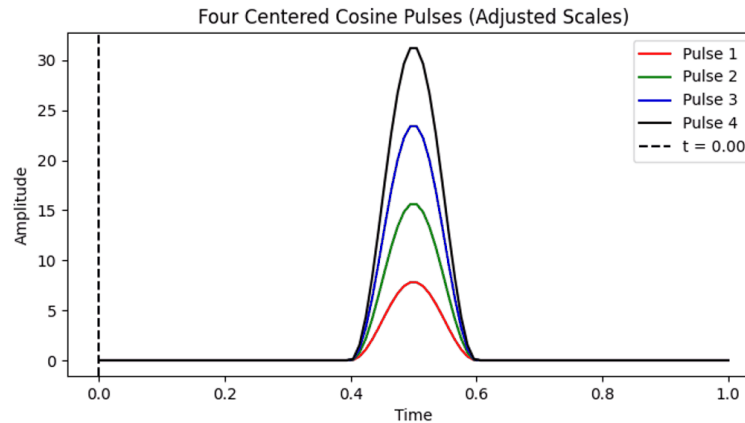
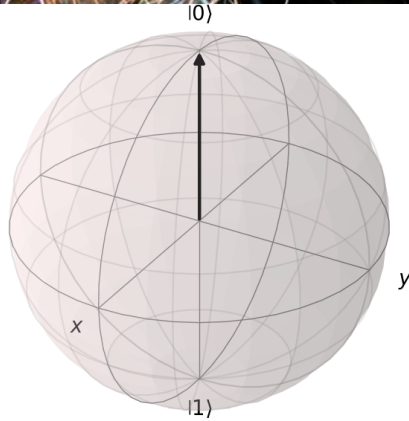
$$V_s(t) = V_0 * f(t)$$



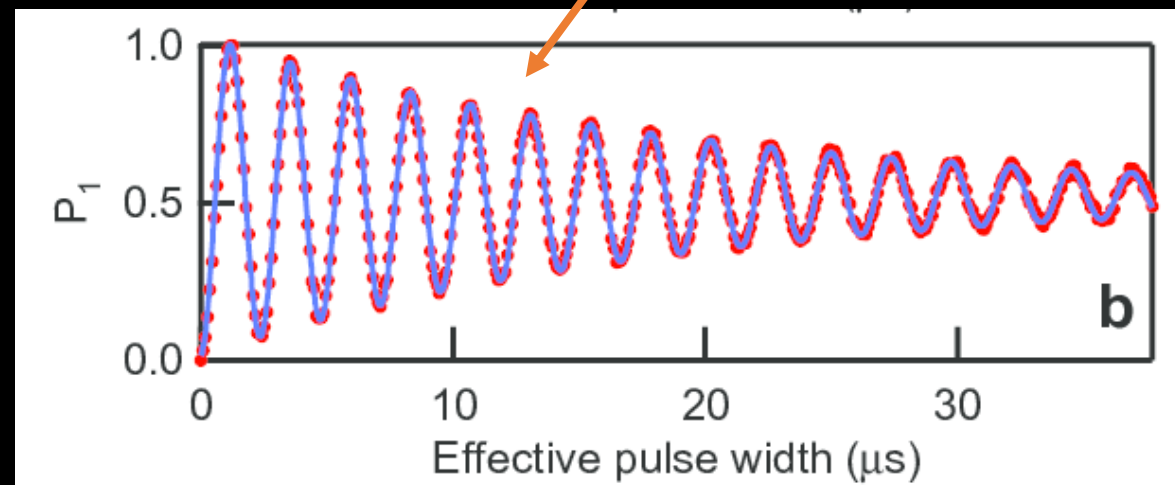
Driving Frequency



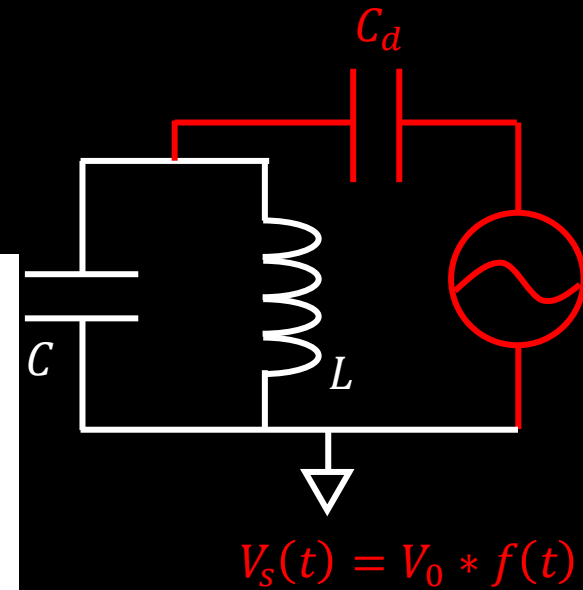
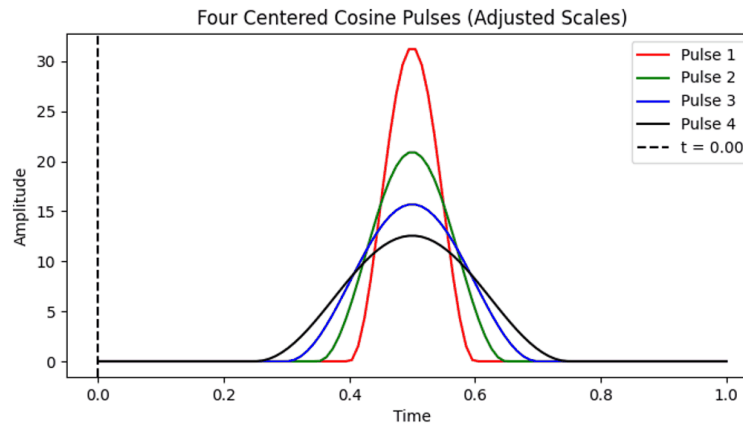
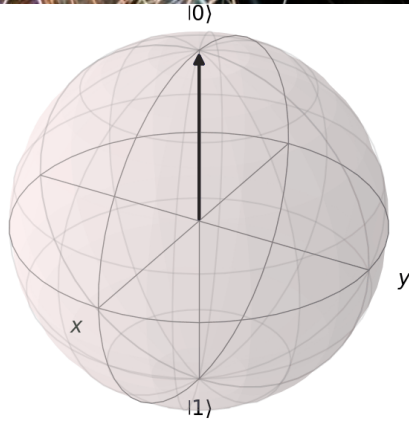
Pulse Calibration



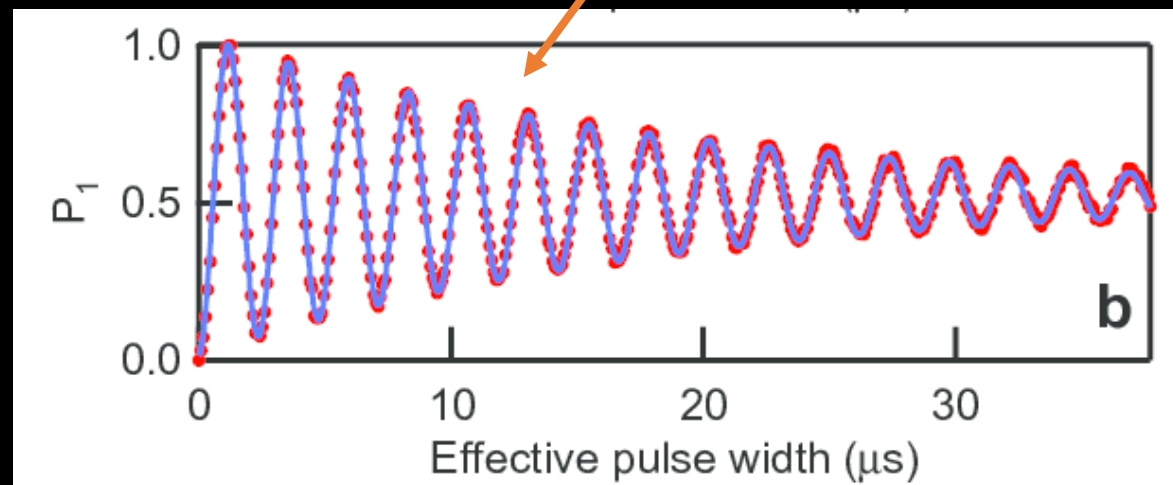
$$\text{Rabi Frequency } \Omega = C_d V_0 \sqrt{\frac{\omega_\Sigma}{2\hbar(C + C_d)}}$$



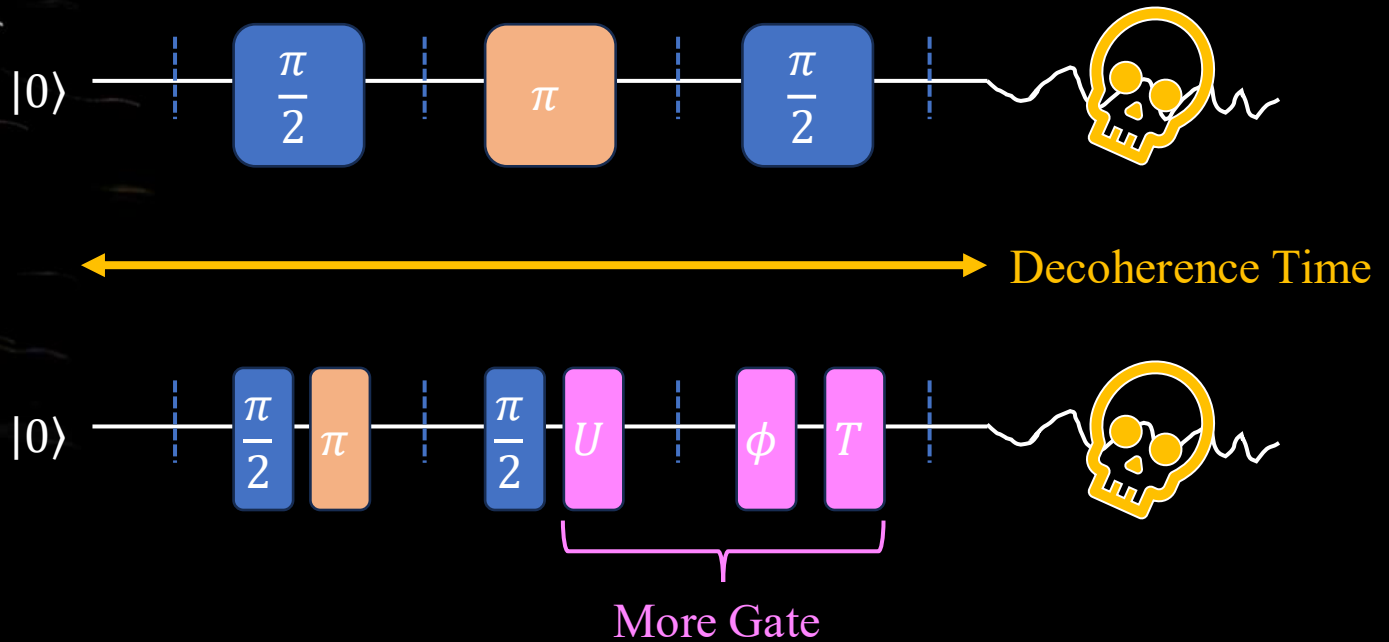
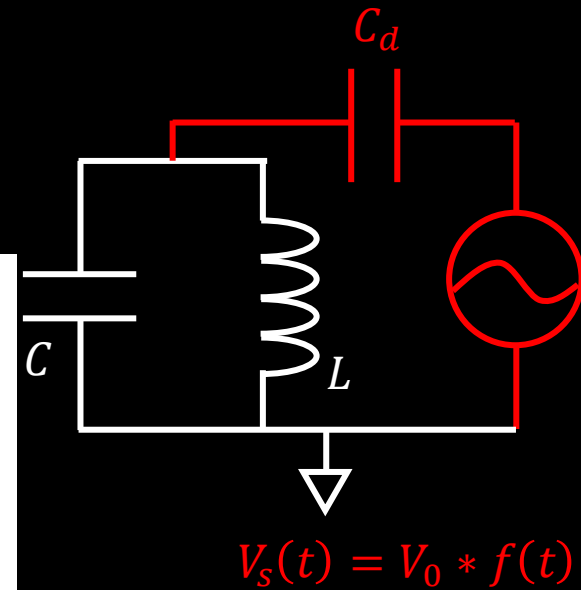
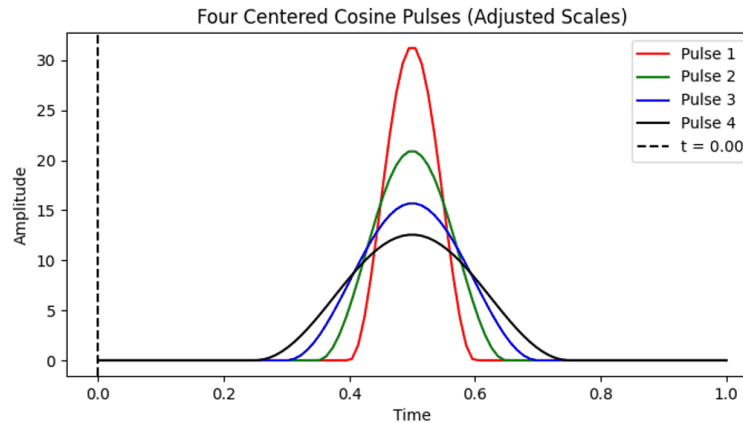
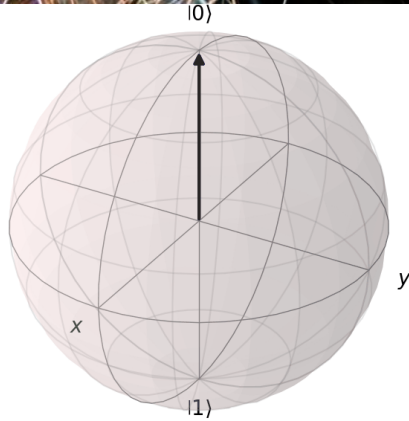
Pulse Calibration



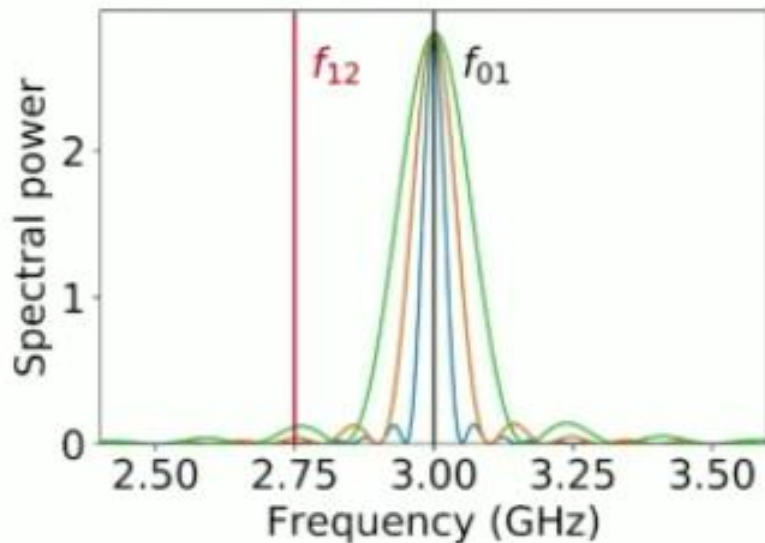
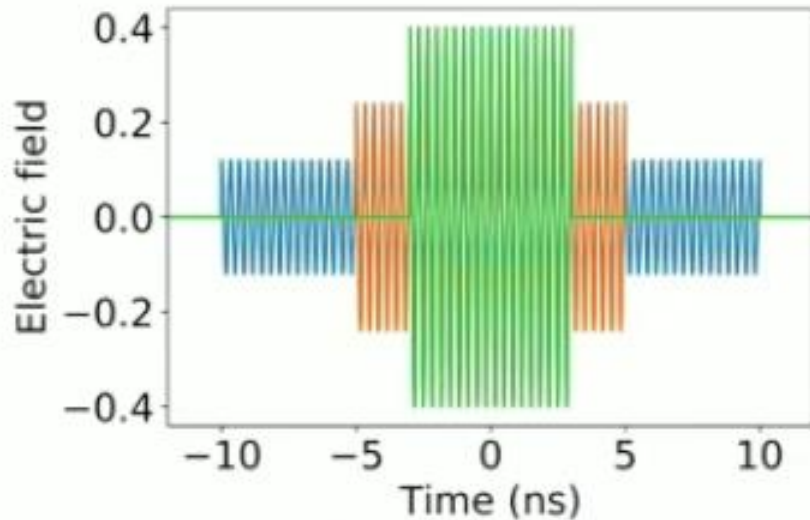
$$\text{Rabi Frequency } \Omega = C_d V_0 \sqrt{\frac{\omega_\Sigma}{2\hbar(C + C_d)}}$$



Pulse Calibration

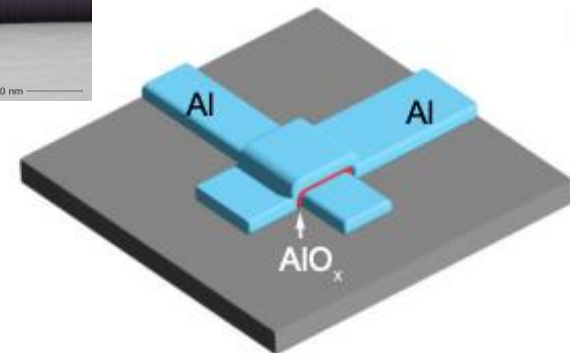
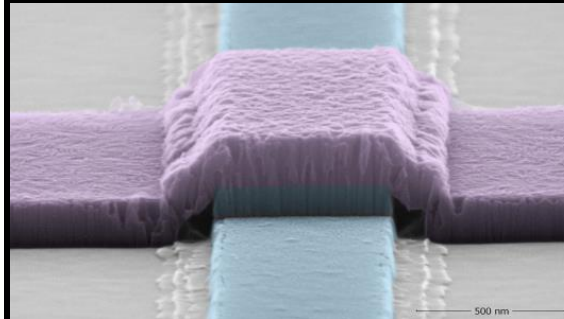


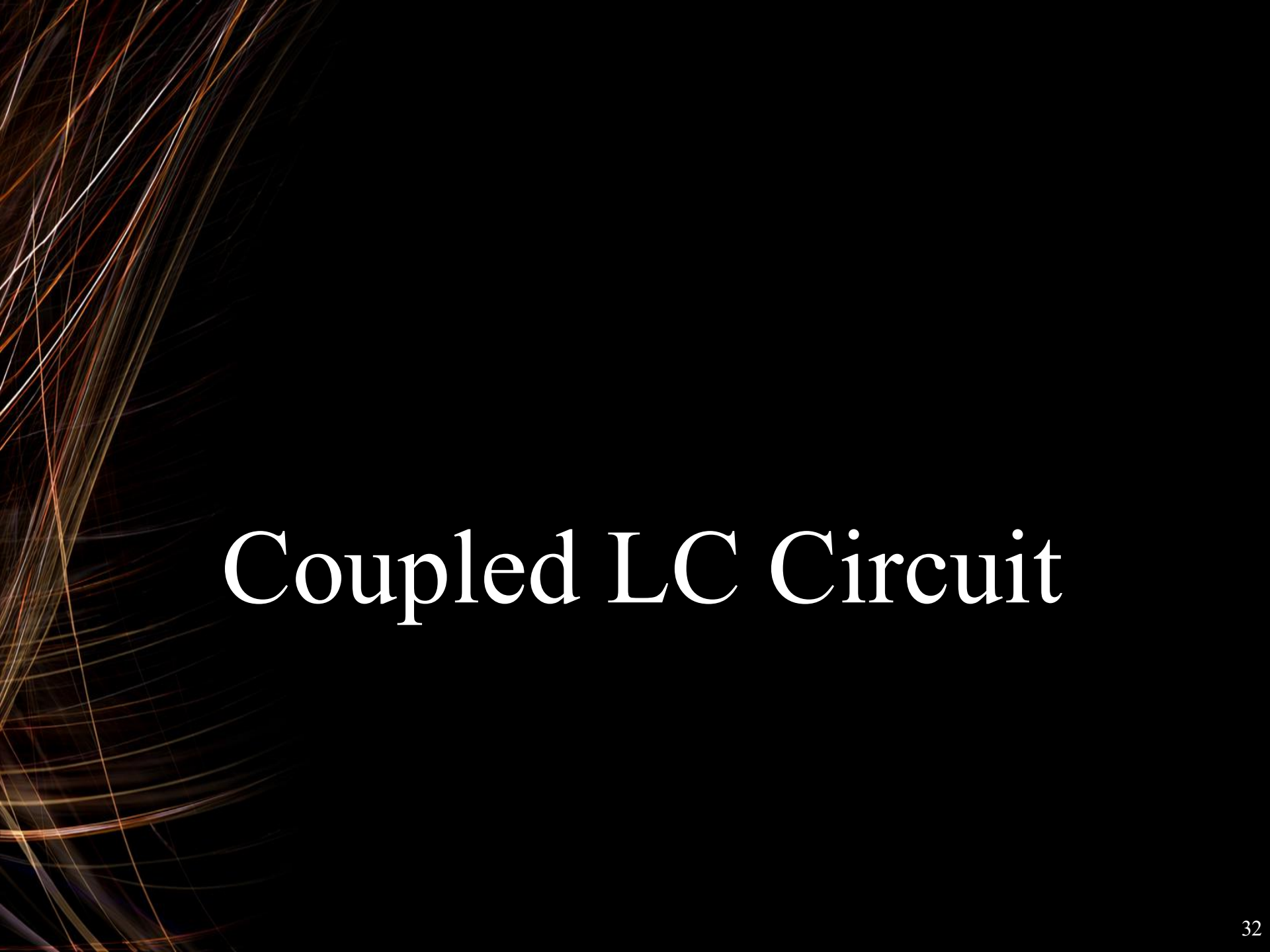
Pulse Calibration



Narrow Pulse

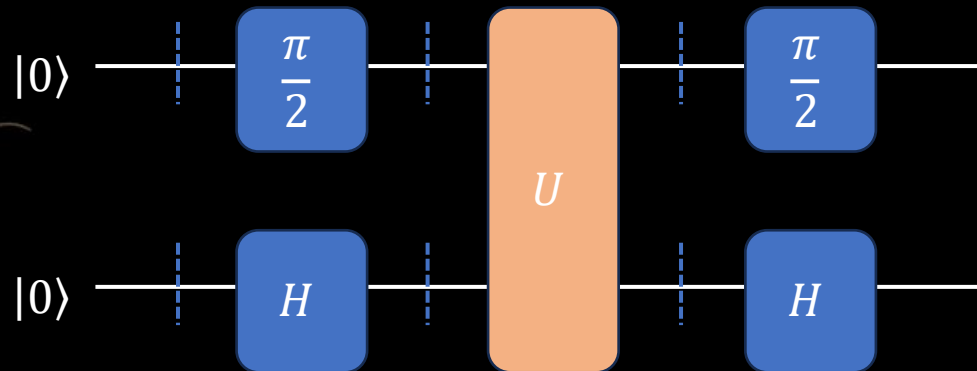
- More Fourier Component
 - $|1\rangle \rightarrow |2\rangle$
 - Higher Frequency Photon
 - Quasi-particle excitation
 - More energy waste $\rightarrow$ Heating
- Junction Change, Qubit Frequency Change, or Qubit Destroy.





Coupled LC Circuit

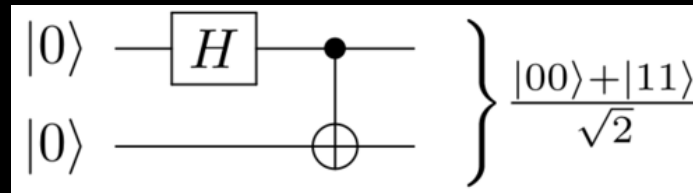
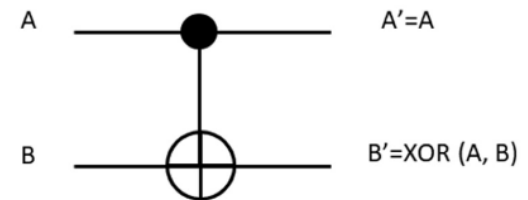
Two Qubit Gate



Logic Table

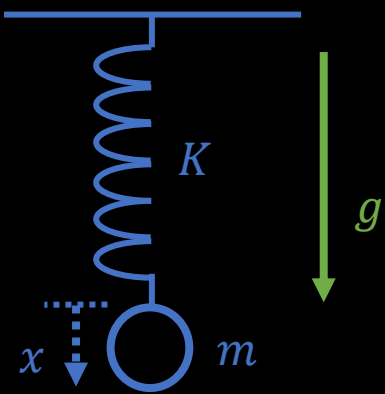
A	B	A'	B'
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

Gate Diagram



$$|0\rangle_A |0\rangle_B \rightarrow \frac{(|0\rangle_A + |1\rangle_B)}{\sqrt{2}} |0\rangle_B \rightarrow \frac{(|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B)}{\sqrt{2}}$$

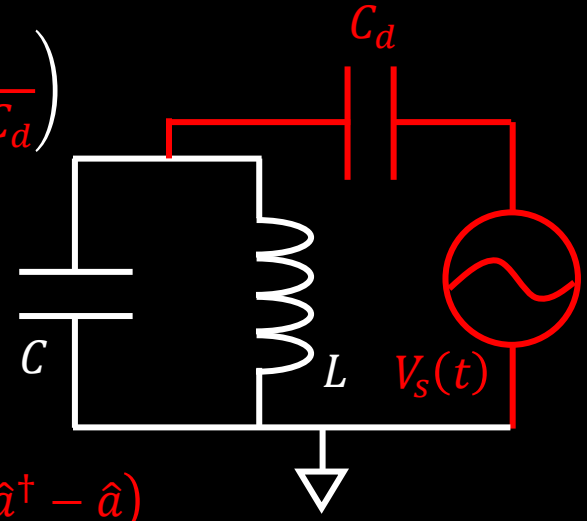
Entanglement/Bell state



Coupled LC

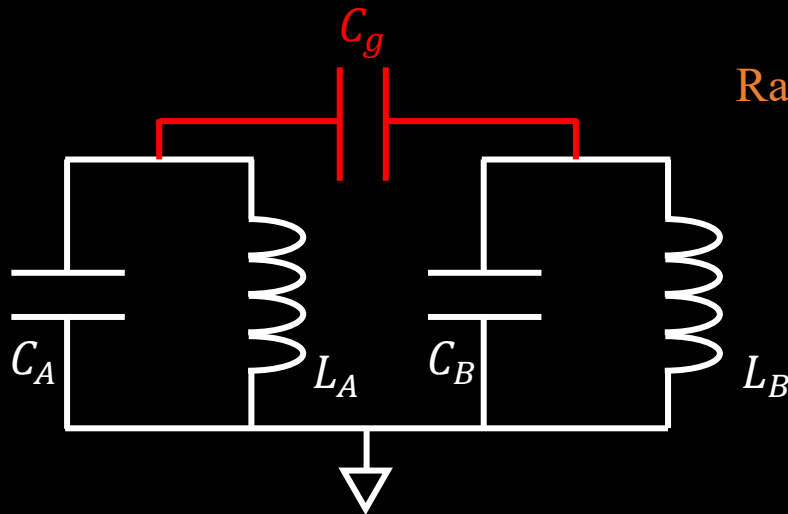
$$C_{\Sigma} = C \left(\frac{C}{C + C_d} \right)$$

$$\hat{H}_d = \frac{Q^2}{2C_{\Sigma}} + \frac{\Phi^2}{2L} - \frac{C_d}{C} V_s(t) Q$$



$$V_s(t) = V_0 * f(t)$$

$$E = \frac{p^2}{2m} + \frac{1}{2} kx^2 - mgx = \hbar\omega_{\Sigma} \left(\hat{a}^{\dagger} \hat{a} + \frac{1}{2} \right) + \hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})$$



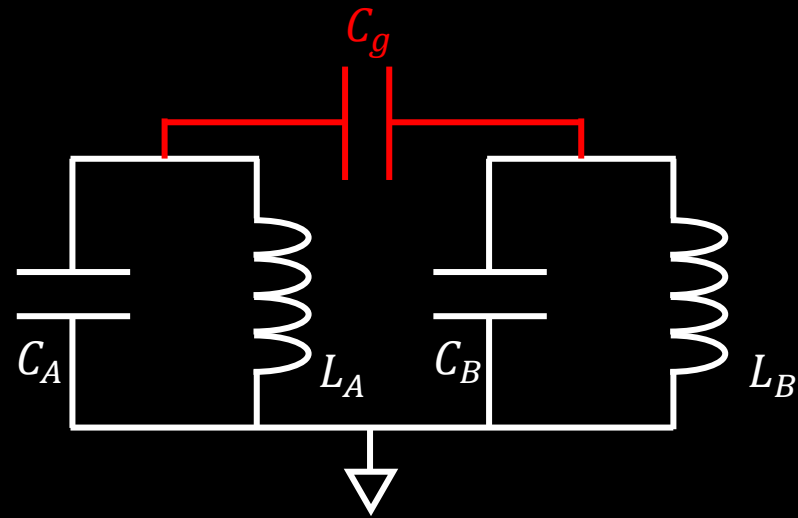
$$\text{Rabi } \Omega = C_d V_0 \sqrt{\frac{\omega_{\Sigma}}{2\hbar(C + C_d)}}$$

$$\begin{cases} \hat{a} \sim (\hat{\Phi} + i\hat{Q}) \\ \hat{a}^{\dagger} \sim (\hat{\Phi} - i\hat{Q}) \end{cases} \leftrightarrow \begin{cases} \Phi \sim (\hat{a}^{\dagger} + \hat{a}) \\ Q \sim -i(\hat{a}^{\dagger} - \hat{a}) \end{cases}$$

$$\begin{aligned} \hat{H}_{2LC} &= \frac{Q_A^2}{2C_{\Sigma A}} + \frac{L_A \Phi_A^2}{2} + \frac{Q_B^2}{2C_{\Sigma B}} + \frac{L_B \Phi_B^2}{2} - C_g \frac{Q_A}{C_A} \frac{Q_B}{C_B} \\ &= \hbar\omega_{\Sigma A} \left(a_{\Sigma A}^{\dagger} a_{\Sigma A} + \frac{1}{2} \right) + \hbar\omega_{\Sigma B} \left(a_{\Sigma B}^{\dagger} a_{\Sigma B} + \frac{1}{2} \right) \\ &\quad - \hbar\Omega_Q i(a_{\Sigma A}^{\dagger} - a_{\Sigma A}) * i(a_{\Sigma B}^{\dagger} - a_{\Sigma B}) \end{aligned}$$

Coupled LC

$$\hat{H}_{2LC} = \frac{Q_A^2}{2C_{\Sigma A}} + \frac{L_A \Phi_A^2}{2} + \frac{Q_B^2}{2C_{\Sigma B}} + \frac{L_B \Phi_B^2}{2} - C_g \frac{Q_A}{C_A} \frac{Q_B}{C_B}$$



$$= \underbrace{\hbar\omega_{\Sigma A} \left(a_{\Sigma A}^\dagger a_{\Sigma A} + \frac{1}{2} \right) + \hbar\omega_{\Sigma B} \left(a_{\Sigma B}^\dagger a_{\Sigma B} + \frac{1}{2} \right)}_{\hat{H}_0} - \underbrace{\hbar\Omega_g i (a_{\Sigma A}^\dagger - a_{\Sigma A}) * i (a_{\Sigma B}^\dagger - a_{\Sigma B})}_{\hat{V}(t)}$$

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = [\hat{H}_0 + \hat{V}(t)] |\psi(t)\rangle$$

$$|\Theta(t)\rangle \equiv U_0^\dagger(t, 0) |\psi(t)\rangle$$

$$U_0^\dagger(t, 0) = \left[\exp \left(\frac{i}{\hbar} T \left[\int_0^t \hat{H}_0(s) ds \right] \right) \right]$$

$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = \hat{H}_r |\Theta(t)\rangle$$

$$\hat{H}_r = U_0^\dagger * \hat{V}(t) * U_0$$

$$= \hbar\Omega_g (e^{i\omega_{\Sigma A} t} a_{\Sigma A}^\dagger - e^{-i\omega_{\Sigma A} t} a_{\Sigma A}) (e^{i\omega_{\Sigma B} t} a_{\Sigma B}^\dagger - e^{-i\omega_{\Sigma B} t} a_{\Sigma B}) |\Theta(t)\rangle$$

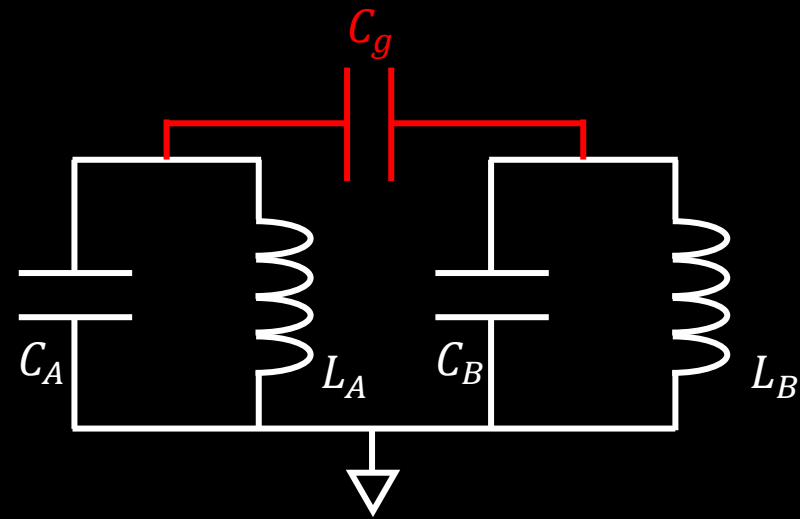
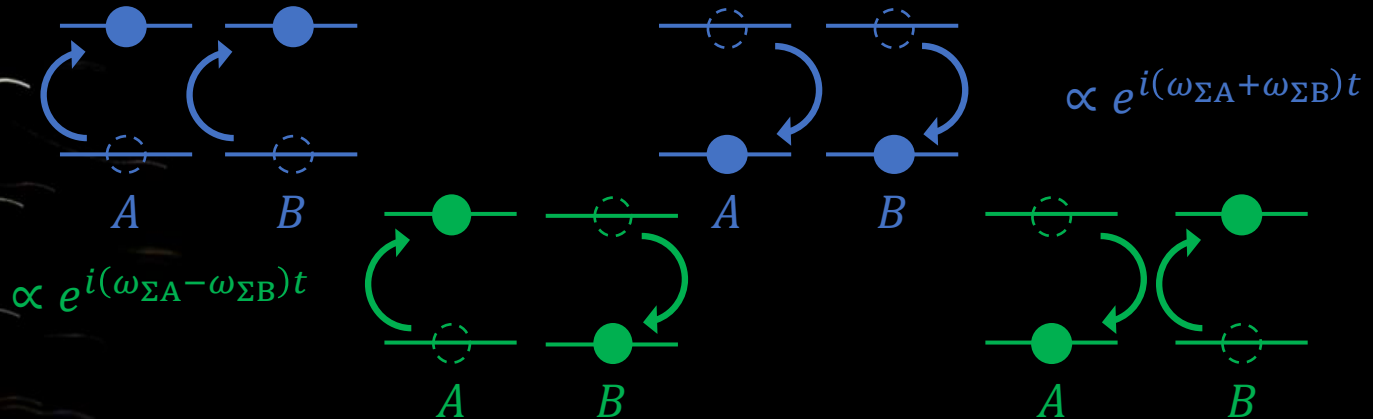
$$= \hbar\Omega_g [e^{i(\omega_{\Sigma A} + \omega_{\Sigma B})t} a_{\Sigma A}^\dagger a_{\Sigma B}^\dagger + e^{-i(\omega_{\Sigma A} + \omega_{\Sigma B})t} a_{\Sigma A} a_{\Sigma B} - e^{i(\omega_{\Sigma A} - \omega_{\Sigma B})t} a_{\Sigma A}^\dagger a_{\Sigma B} - e^{-i(\omega_{\Sigma A} - \omega_{\Sigma B})t} a_{\Sigma A} a_{\Sigma B}^\dagger] |\Theta(t)\rangle$$

Coupled LC

$$\hat{H}_r = U_0^\dagger * \hat{V}(t) * U_0$$

$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = \hat{H}_r |\Theta(t)\rangle$$

$$= \hbar\Omega_g \left[\underbrace{e^{i(\omega_{\Sigma A} + \omega_{\Sigma B})t} a_{\Sigma A}^\dagger a_{\Sigma B}^\dagger}_{\propto e^{i(\omega_{\Sigma A} - \omega_{\Sigma B})t}} + \underbrace{e^{-i(\omega_{\Sigma A} + \omega_{\Sigma B})t} a_{\Sigma A} a_{\Sigma B}}_{\propto e^{i(\omega_{\Sigma A} + \omega_{\Sigma B})t}} - e^{i(\omega_{\Sigma A} - \omega_{\Sigma B})t} a_{\Sigma A}^\dagger a_{\Sigma B} - e^{-i(\omega_{\Sigma A} - \omega_{\Sigma B})t} a_{\Sigma A} a_{\Sigma B}^\dagger \right] |\Theta(t)\rangle$$

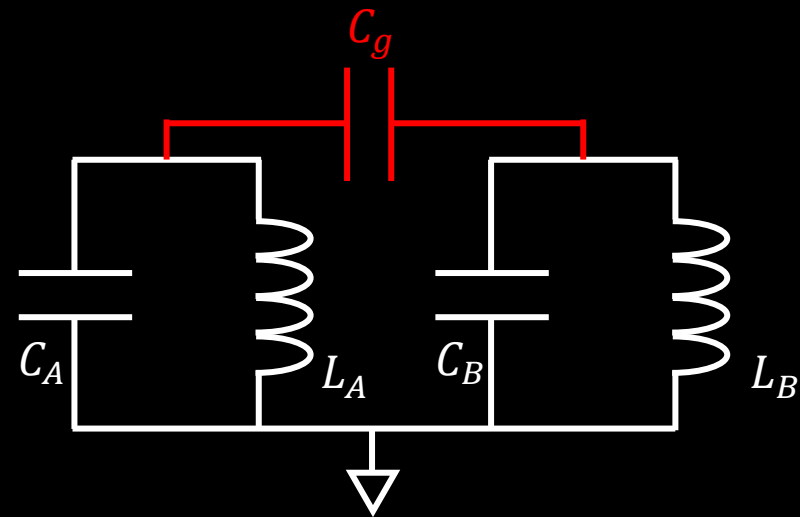
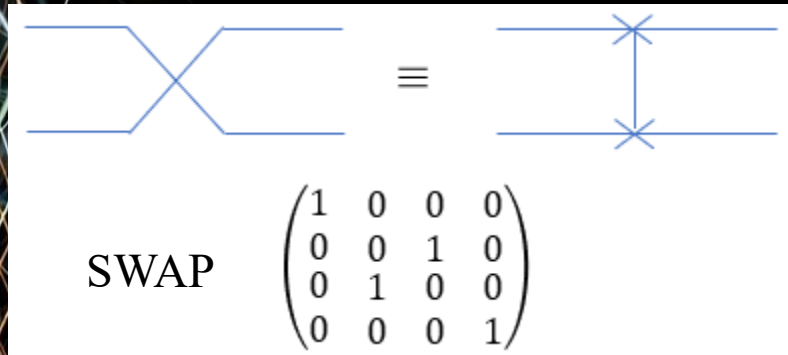


Rotating Wave Approximation (RWA)

1. $\omega \sim \text{GHz}$, detuning $\Delta \equiv \omega_{\Sigma A} - \omega_{\Sigma B} \sim 50 \text{ MHz}$, $2\omega \gg \Delta$ too fast

2. Transition $|i\rangle \rightarrow |f\rangle \propto -\frac{i}{\hbar} \int_0^t \langle f | \hat{H}_r | i \rangle dt \sim \frac{\dots}{\omega_{\Sigma A} + \omega_{\Sigma B}}$ or $\frac{\dots}{\omega_{\Sigma A} - \omega_{\Sigma B}}$

SWAP (iSWAP)



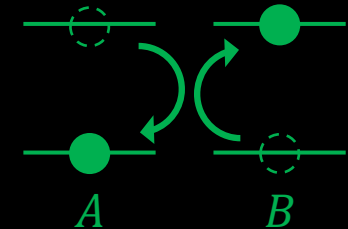
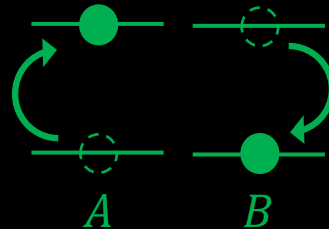
$$|0_A 0_B\rangle \rightarrow |0_A 0_B\rangle$$

$$|1_A 1_B\rangle \rightarrow |1_A 1_B\rangle$$

$$|0_A 1_B\rangle \rightarrow i|1_A 0_B\rangle$$

$$|1_A 0_B\rangle \rightarrow i|0_A 1_B\rangle$$

$$\propto e^{i(\omega_{\Sigma A} - \omega_{\Sigma B})t}$$

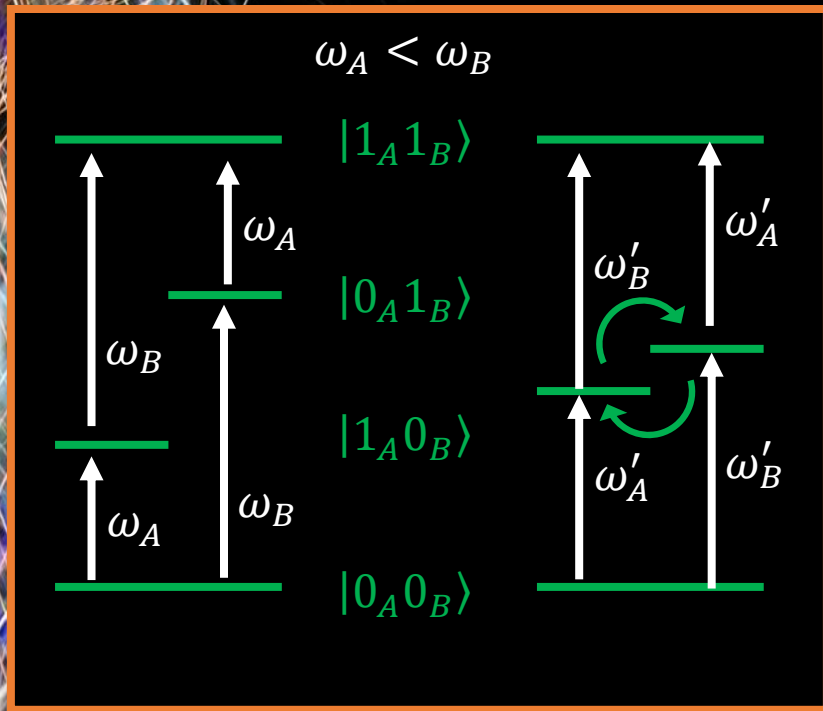


Rotating Wave Approximation (RWA)

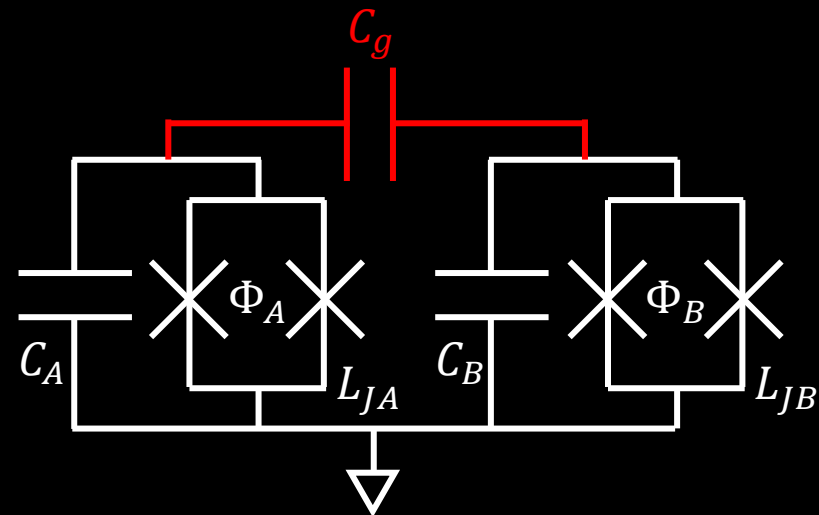
1. $\omega \sim \text{GHz}$, detuning $\Delta \equiv \omega_{\Sigma A} - \omega_{\Sigma B} \sim 50 \text{ MHz}$, $2\omega \gg \Delta$ too fast

2. Transition $|i\rangle \rightarrow |f\rangle \propto -\frac{i}{\hbar} \int_0^t \langle f | \hat{H}_r | i \rangle dt \sim \frac{\dots}{\Delta} = \frac{\dots}{\omega_{\Sigma A} - \omega_{\Sigma B}}$

SWAP (iSWAP)



$$\propto e^{i(\omega_{\Sigma A} - \omega_{\Sigma B})t}$$

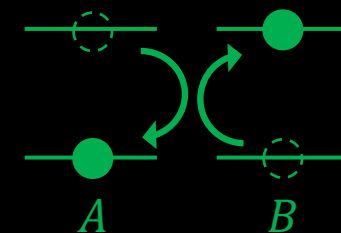
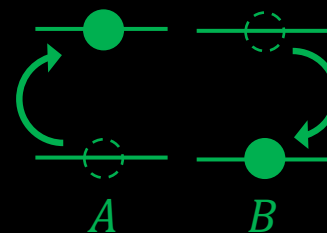


$$|0_A 0_B\rangle \rightarrow |0_A 0_B\rangle$$

$$|1_A 1_B\rangle \rightarrow |1_A 1_B\rangle$$

$$|0_A 1_B\rangle \rightarrow i|1_A 0_B\rangle$$

$$|1_A 0_B\rangle \rightarrow i|0_A 1_B\rangle$$

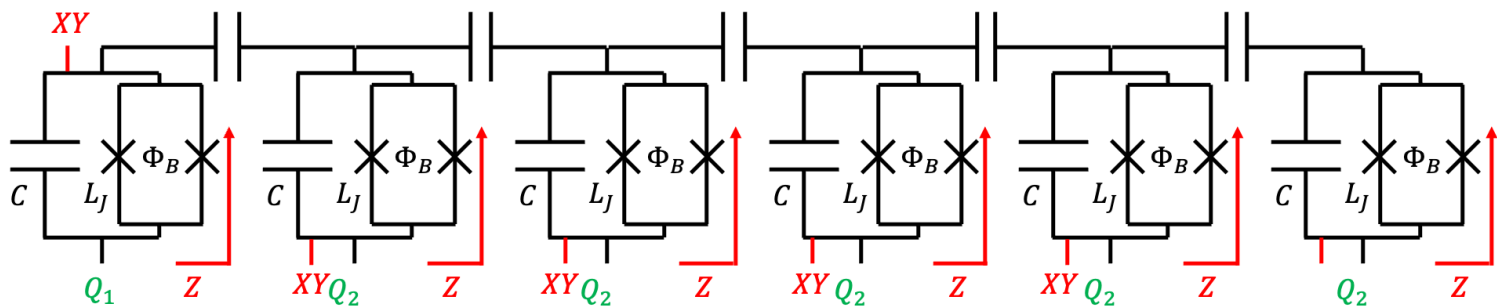
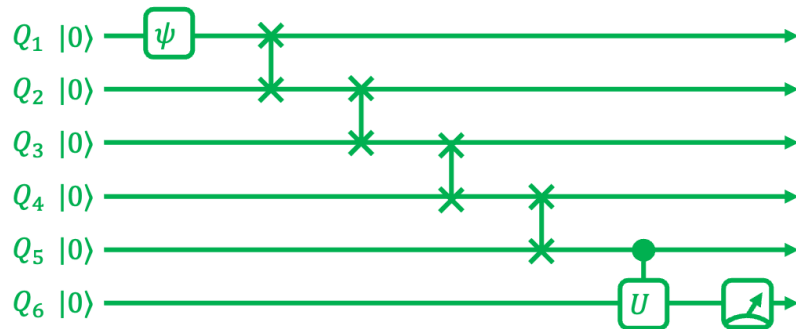


Rotating Wave Approximation (RWA)

1. $\omega \sim \text{GHz}$, detuning $\Delta \equiv \omega_{\Sigma A} - \omega_{\Sigma B} \sim 50 \text{ MHz}$, $2\omega \gg \Delta$ too fast

2. Transition $|i\rangle \rightarrow |f\rangle \propto -\frac{i}{\hbar} \int_0^t \langle f | \hat{H}_r | i \rangle dt \sim \frac{\dots}{\Delta} = \frac{\dots}{\omega_{\Sigma A} - \omega_{\Sigma B}}$

SWAP (iSWAP)



Multi Qubit Circuit

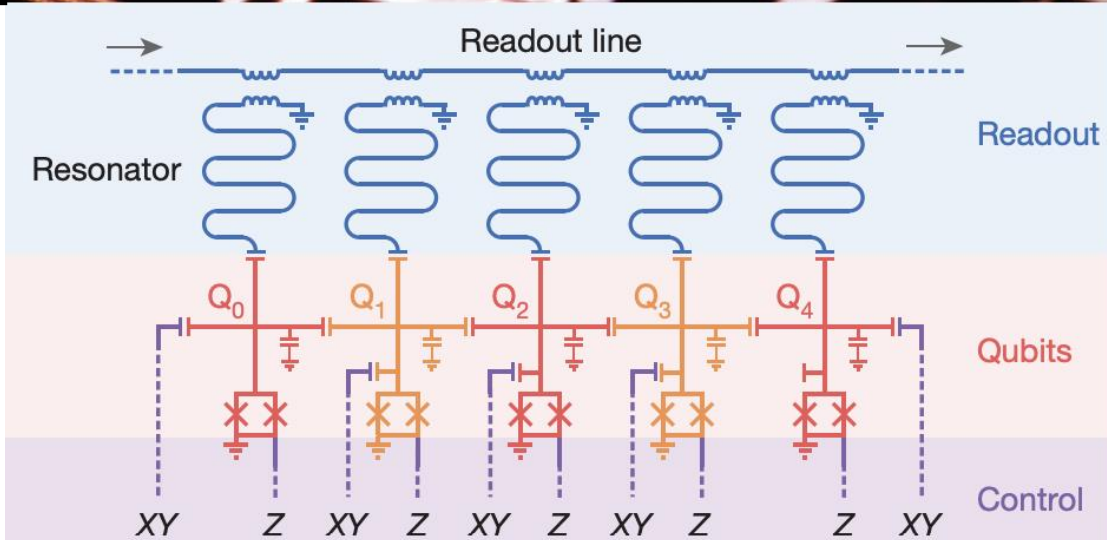
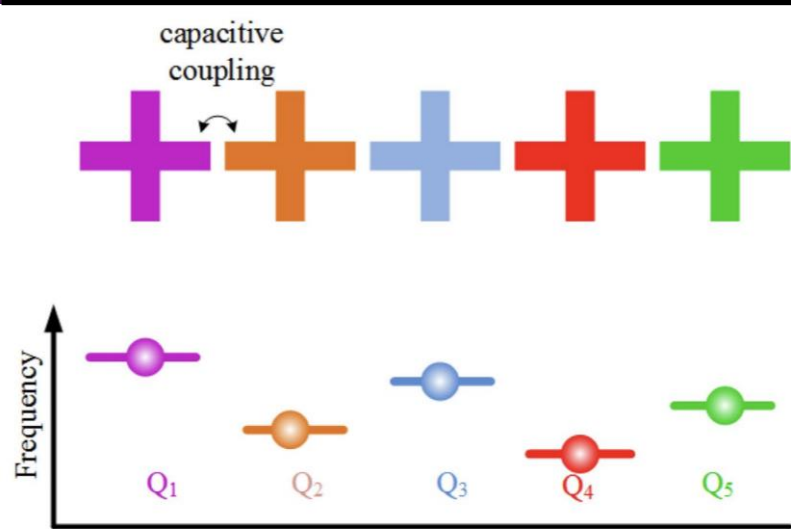
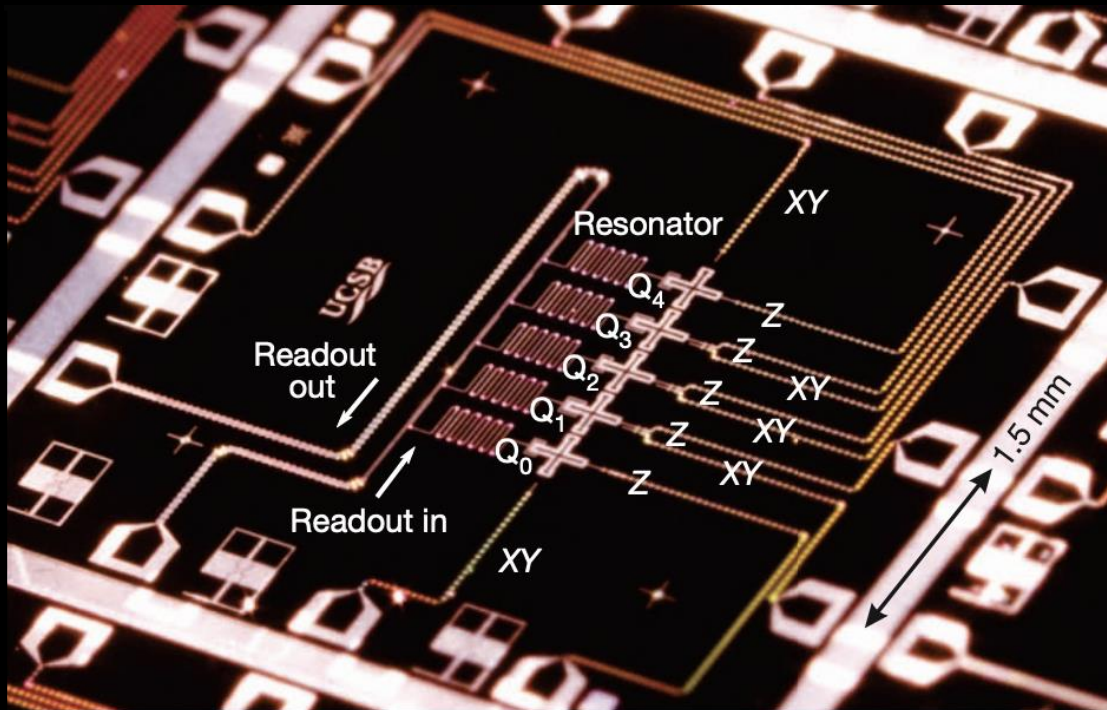
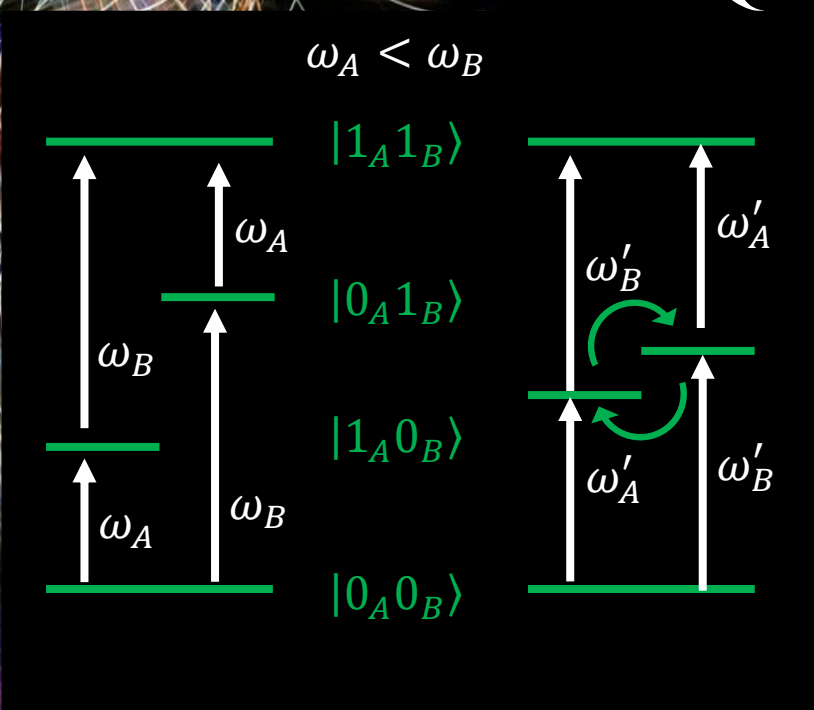
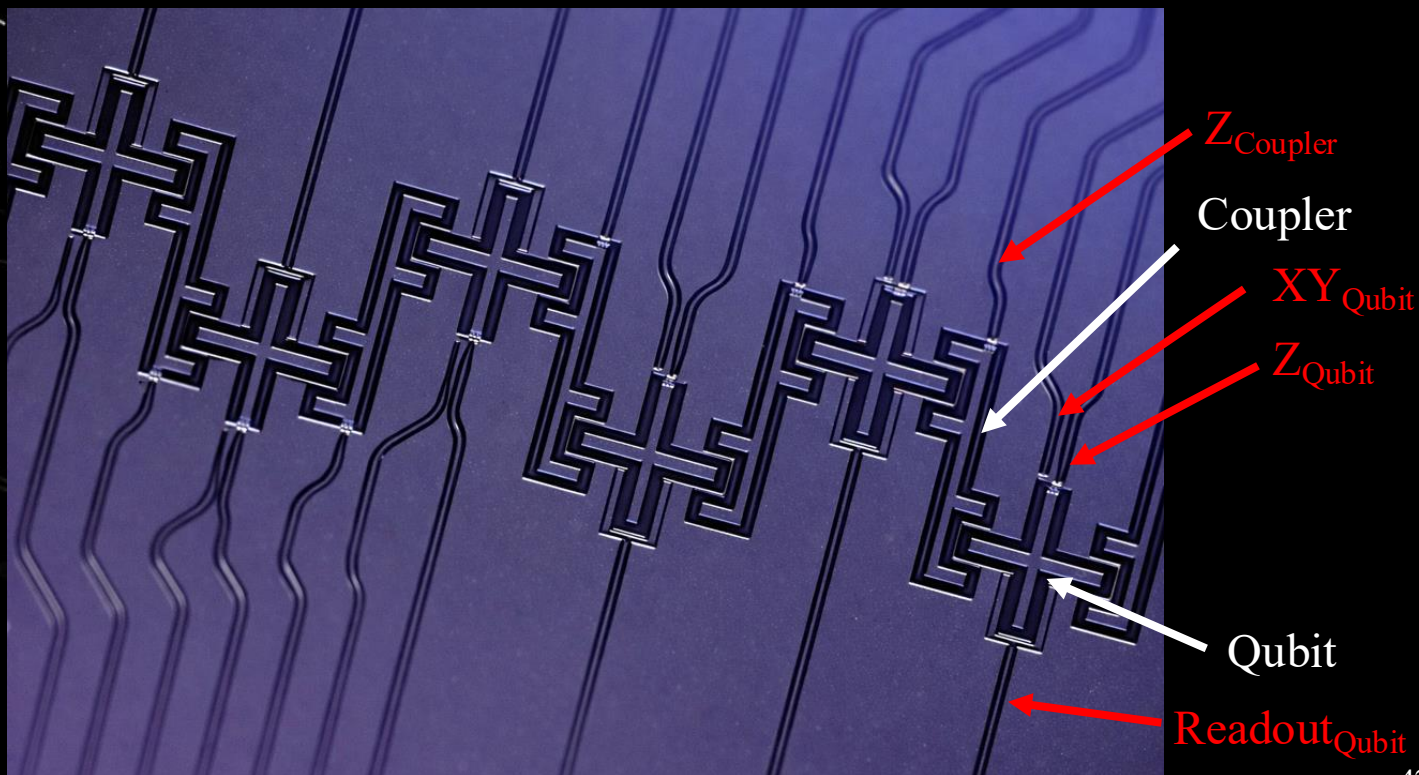
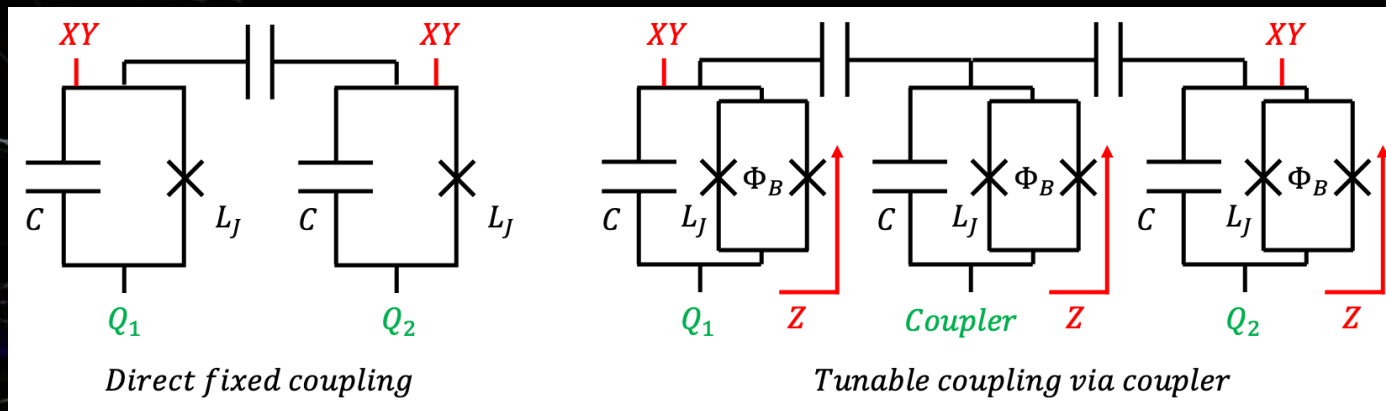


Figure 1 consists of two parts. Part (a) is a schematic diagram of a 2D lattice structure, showing a central node highlighted in red, surrounded by other nodes and edges. Part (b) is a 3D visualization of the lattice structure, showing a dense network of colored lines representing the connections between nodes.

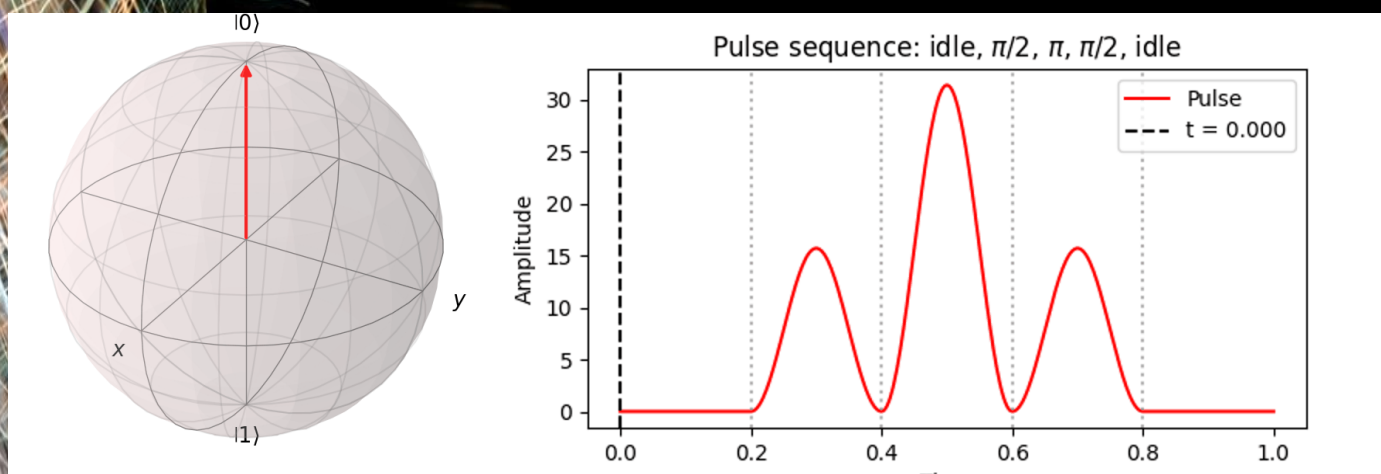
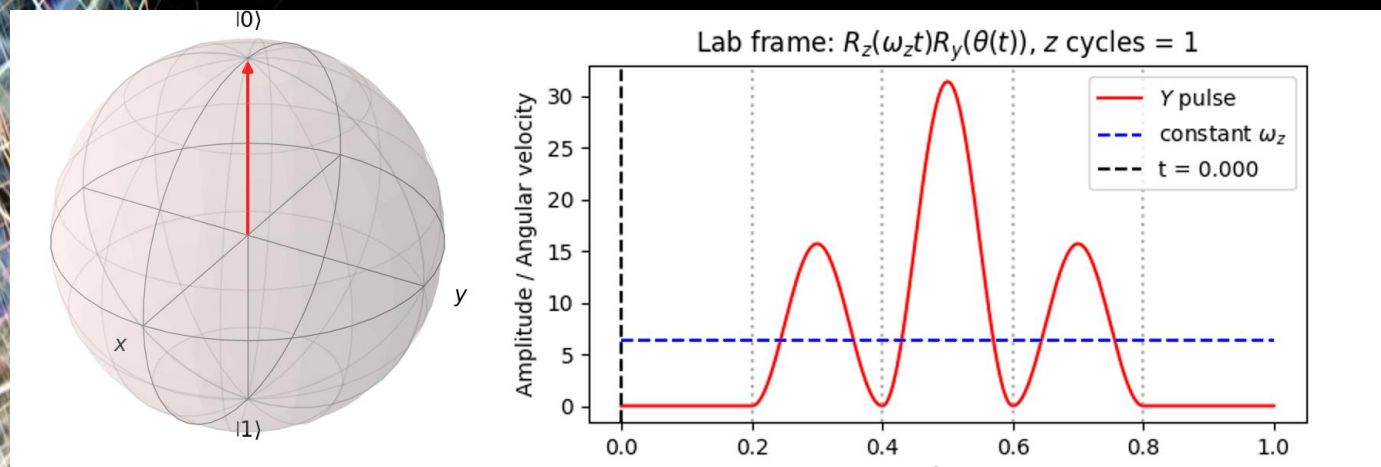
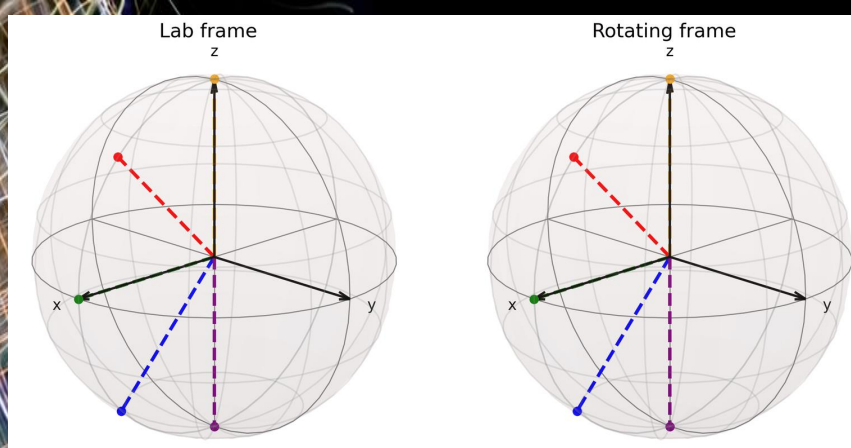
(a) Grid Topology (b) IBM's Heavy-hexagonal (c) Google's Sycamore

Tunable Coupler

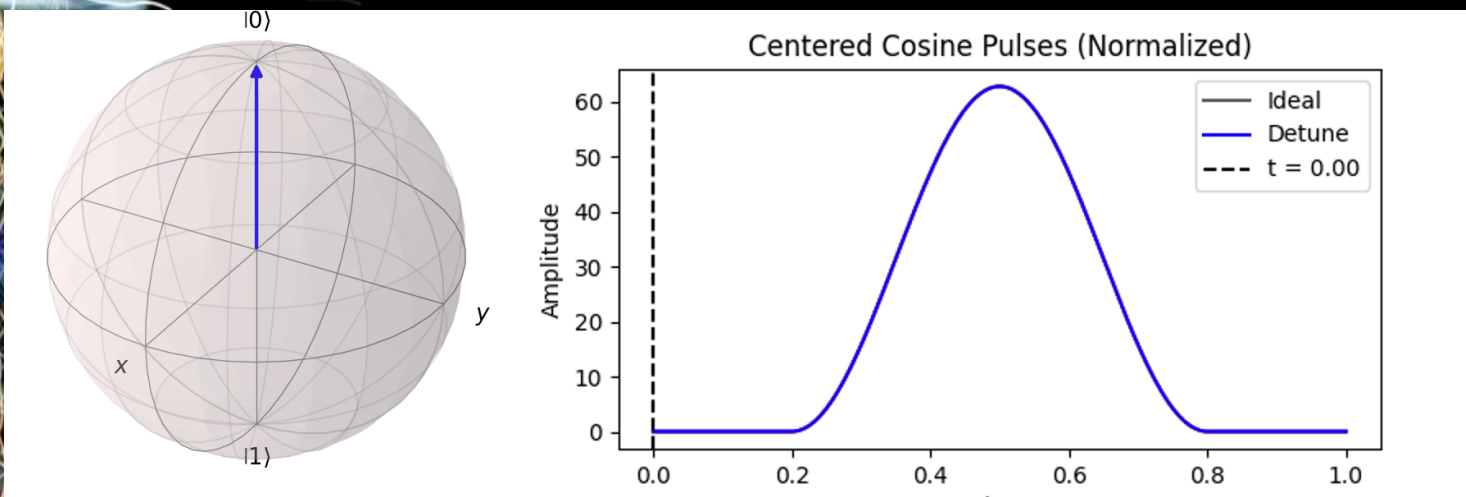
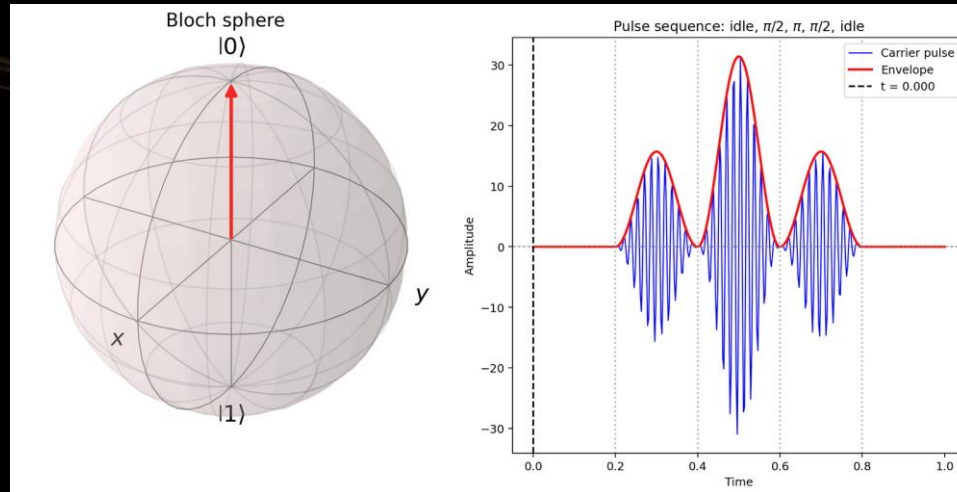




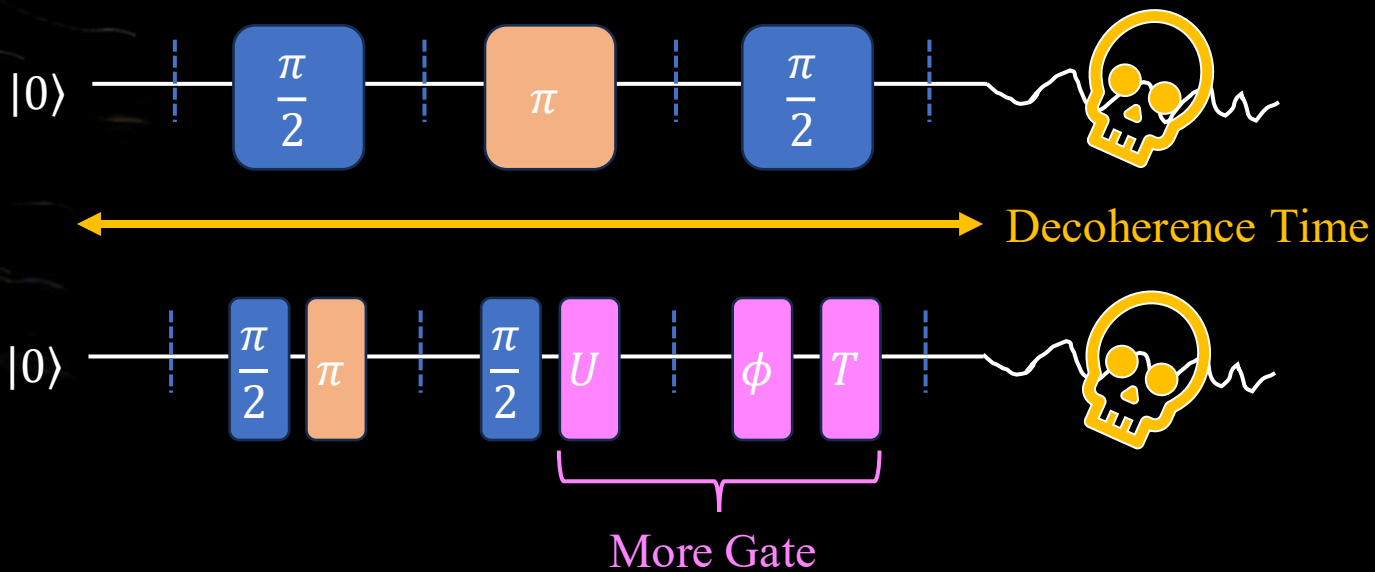
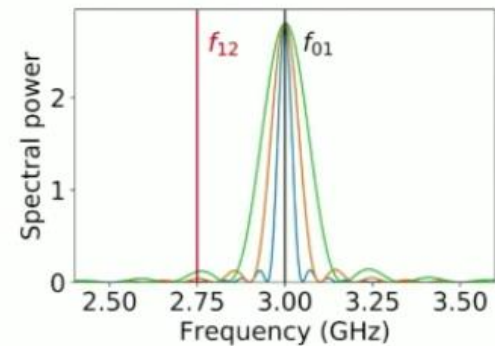
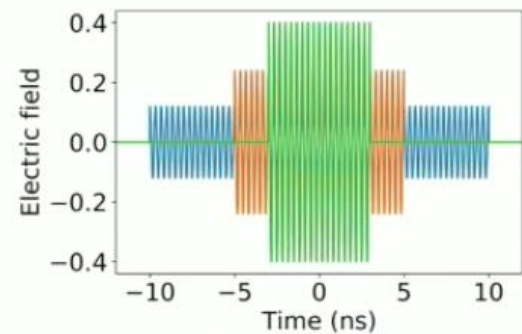
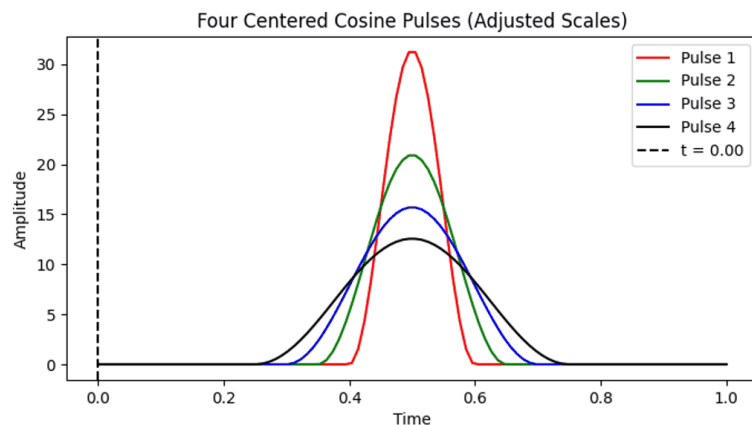
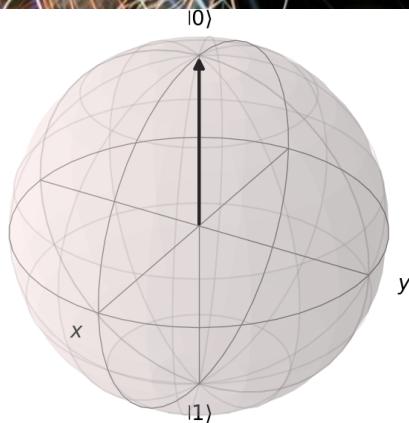
Summary



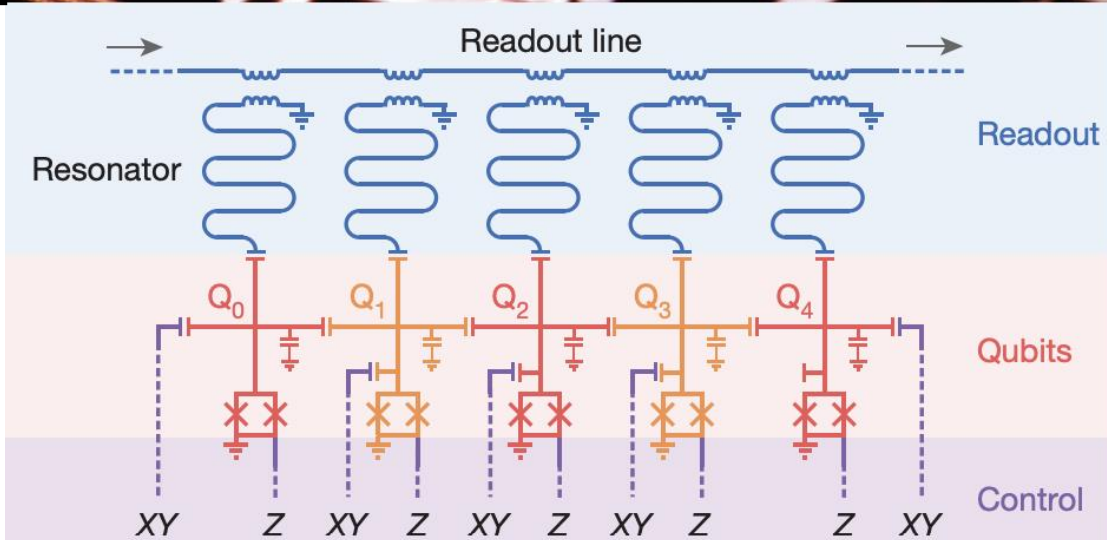
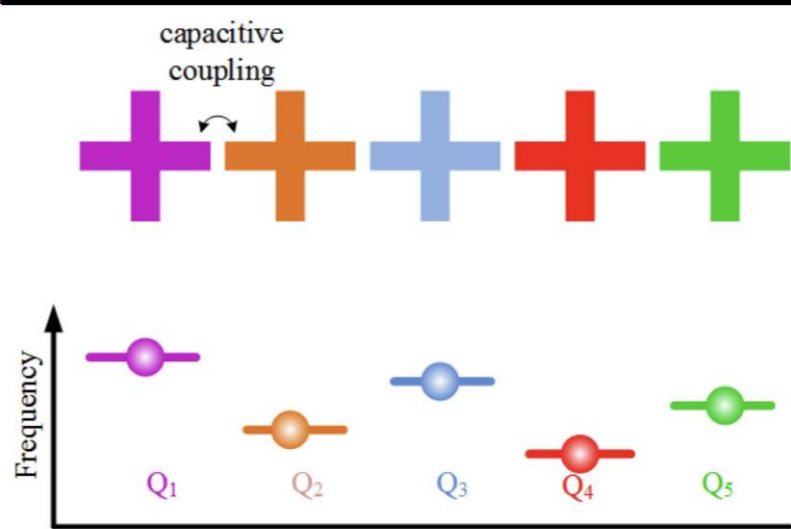
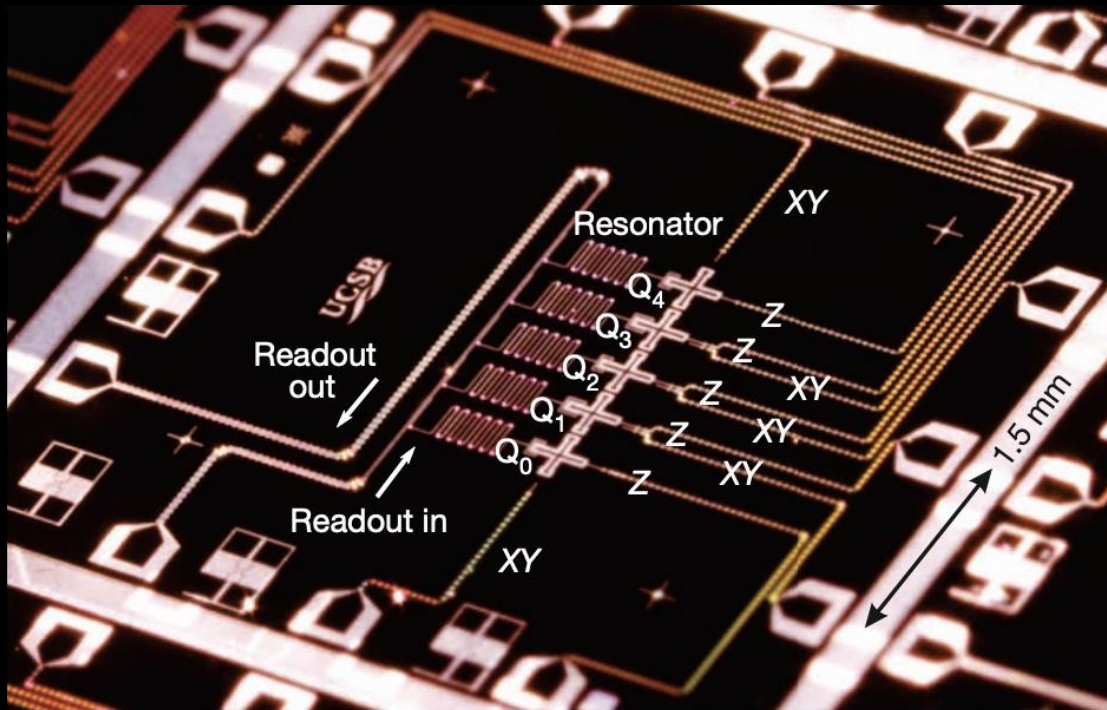
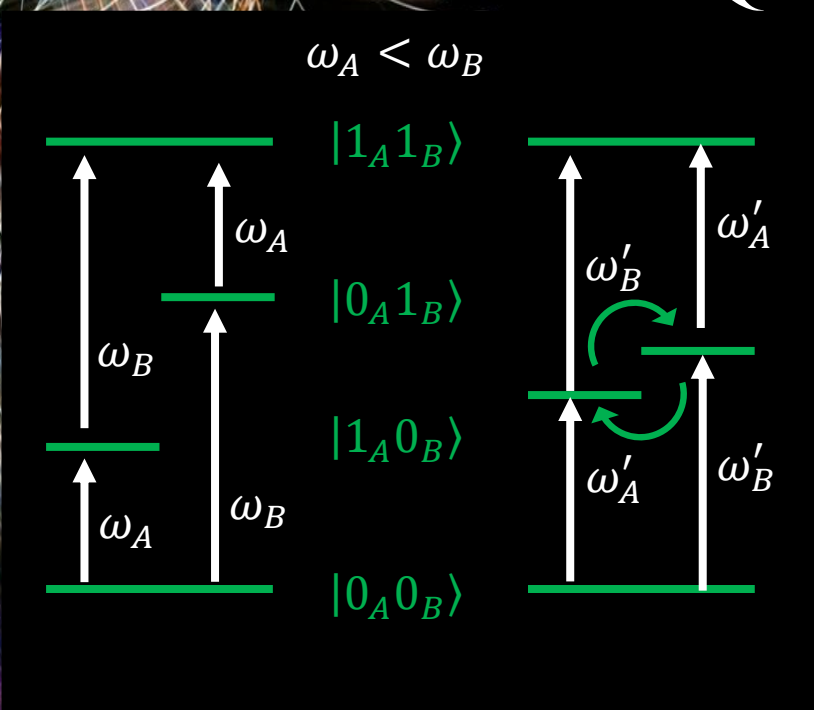
Driving Frequency



Pulse Calibration



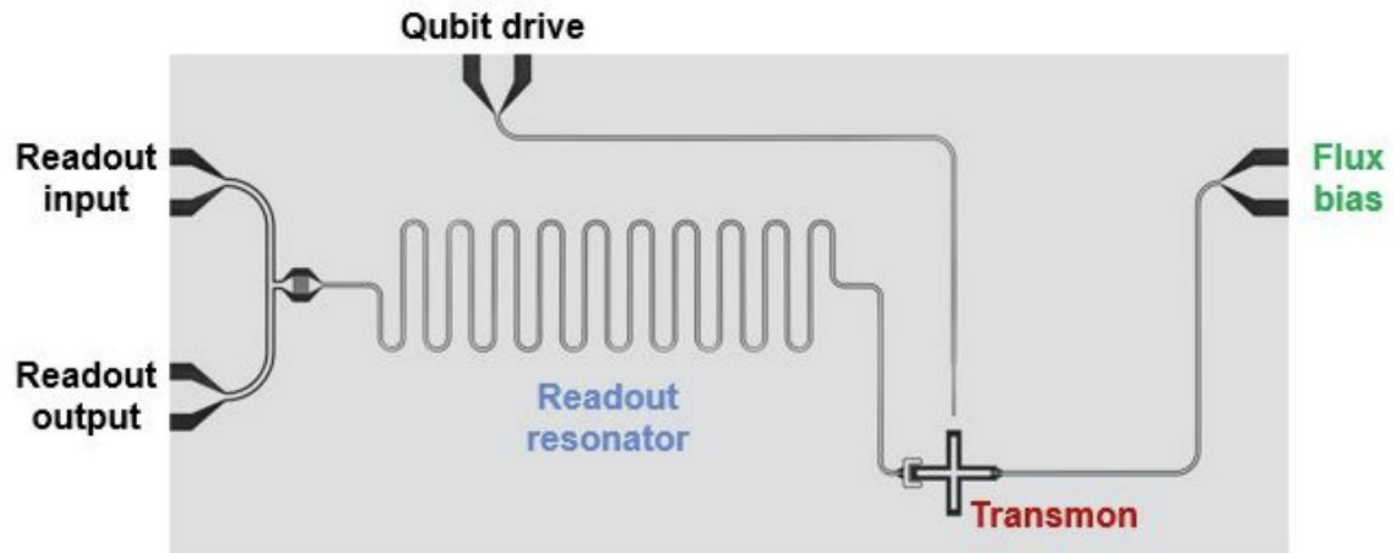
Multi Qubit Circuit



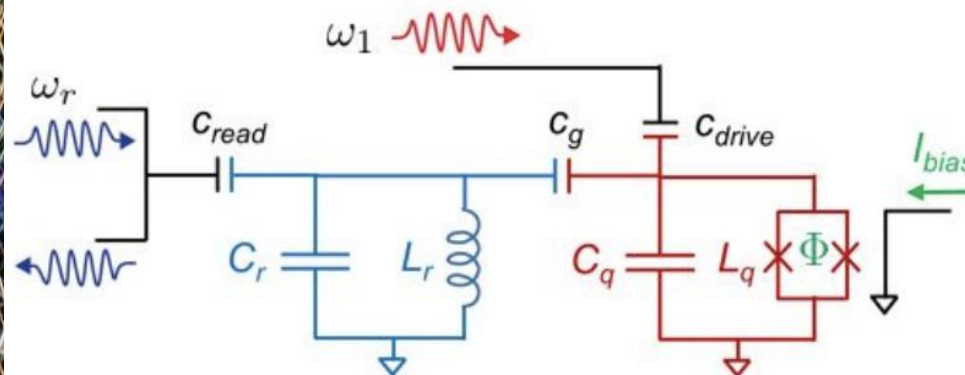


Next week

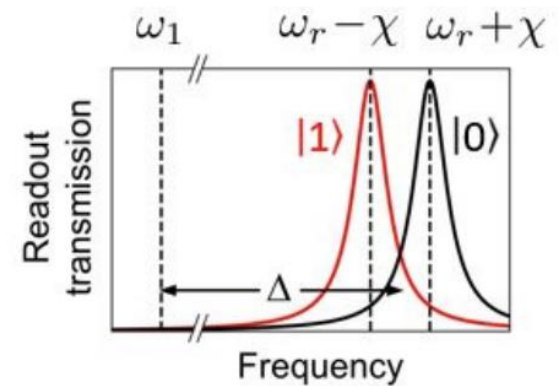
Readout Circuit



(a)



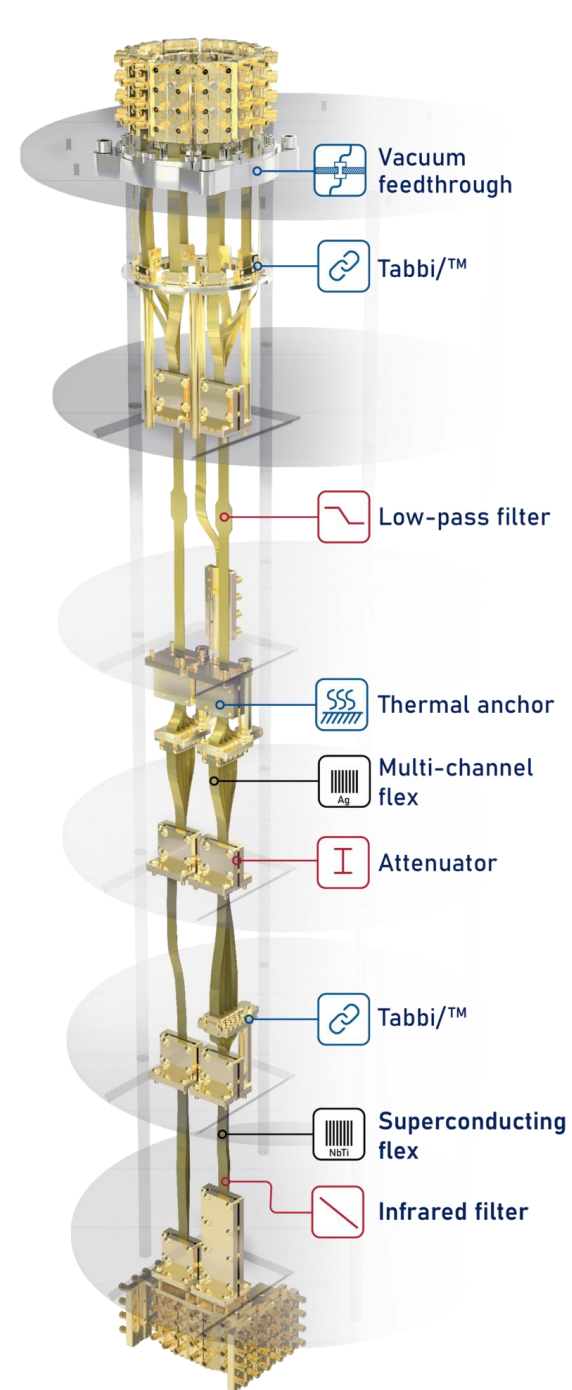
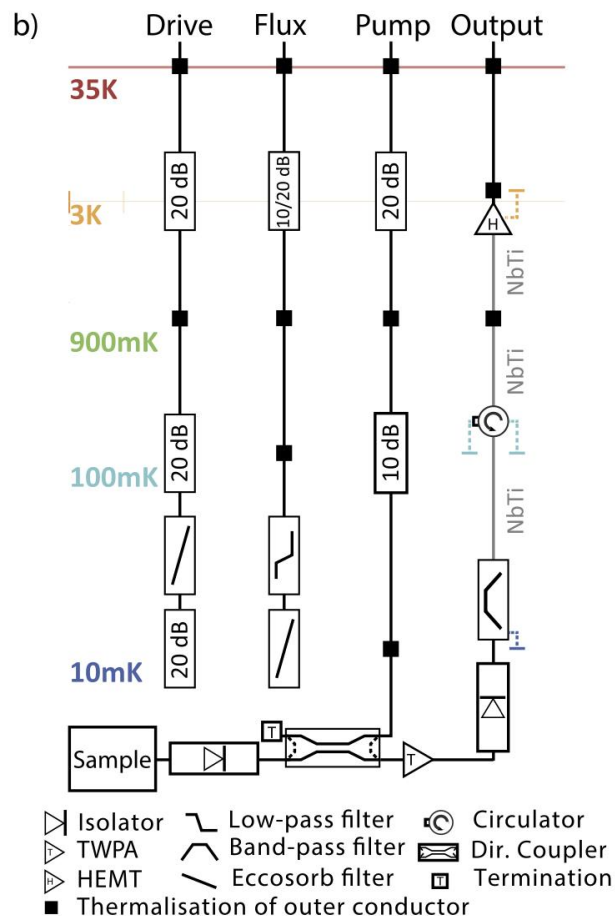
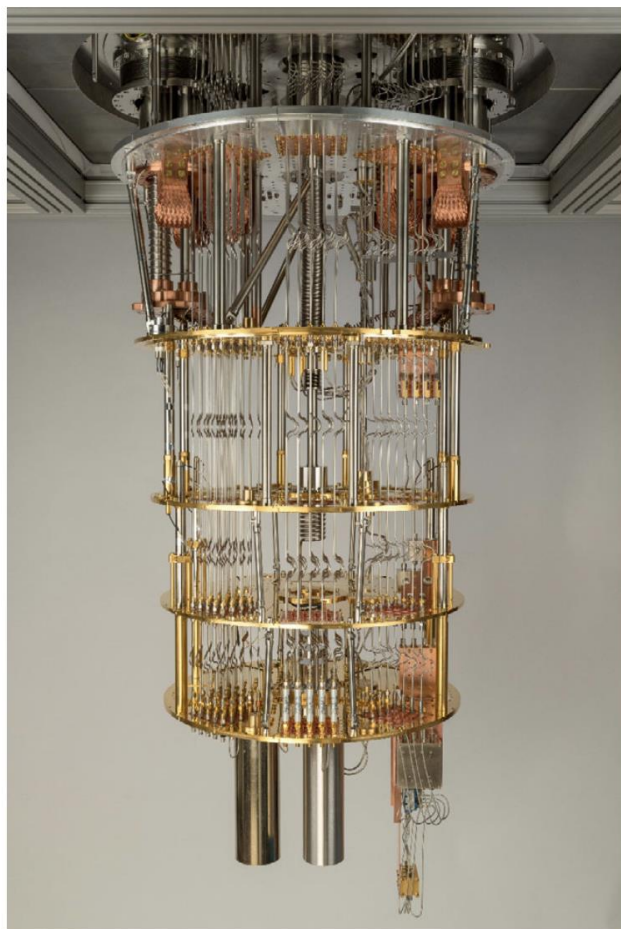
(b)



(c)



a)





Thank you!